AXKU062 Development Board User Manual

Alinx Electronic Limited

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Document Revision History

chapter	Revision Overview
V1.0	
all	First version released
2025-10-14 V2.0	
1.9 Connector Pin Definitions	PAGE23 Serial configuration signals are connected to the board-to-board connector.
2026-1-13 V3.0	
1.3 DDR4 DRAM 1.2 FPGA Chip	1、DDR chip modification: Replaced the original MT40A512M16LY-062EIT with Nanya's NT5AD512M16C4-JRI. 2、GTH has been changed from 12.5Gb/s to 16.3Gb/s

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Kintex UltraScale-based development platform (model:AXKU062) from Alinx Electronic Limited has been officially released. To help you quickly understand this development platform, we have edit this user manual.

The AXKU062 adopts a core board plus expansion board design, facilitating secondary development and utilization of the core board by users. The core board is equipped with four 1GB high-speed DDR4 SDRAM chips and two 128Mb QSPI FLASH chips.

The expansion board design provides users with a wealth of interfaces: two 10G SFP+ fiber optic interfaces, three FMC expansion interfaces (one HPC, two LPC), one gigabit Ethernet port, one UART serial port, one SD card interface, LED buttons, etc.The image below shows the actual development board.

Figure 1.1 - Physical image of the development board

The following diagram shows the overall structure of the development system:

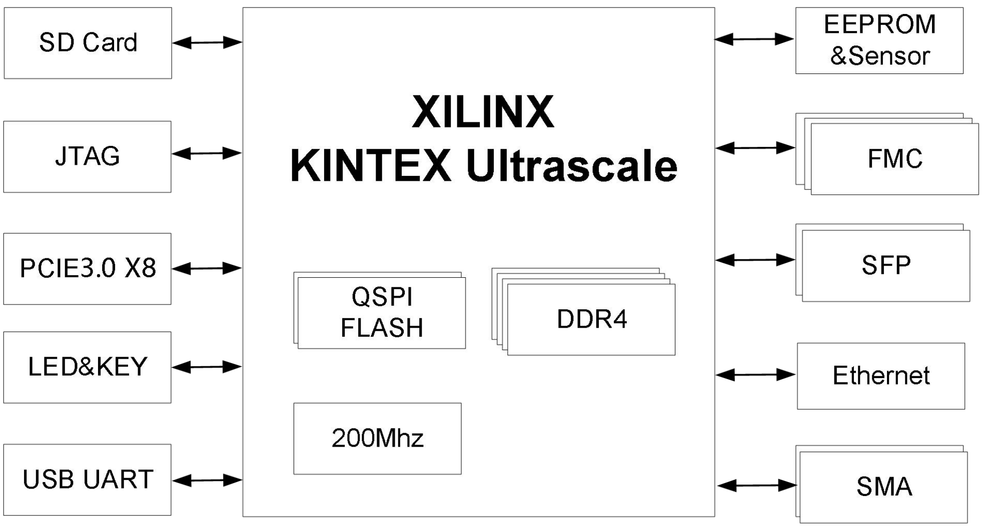

Figure 1.2 - Schematic diagram of the development system

This diagram shows the interfaces and functions that our development platform can contain.

FPGA core board

- FPGA chip: Xilinx Kintex UltraScale chip XCKU060 .
- DDR4: Features four high-capacity 1GB (4GB total) high-speed DDR4 SDRAM chips. Can be used for FPGA data storage, image analysis caching, and data processing.
- QSPI FLASH: Two 128Mbit QSPI NOR FLASH memory chips, which can be used as storage for FPGA chip configuration files and user data;
- One 200MHz differential crystal oscillator.
- Two light-emitting diodes (LEDs), one power indicator, and one DONE configuration indicator.

Expansion board

- It has two SFP+ fiber optic interfaces, each with a fiber optic data communication reception and transmission speed of up to 16.3Gb/s.
- One PCIe 3.0 x8 interface, endpoint mode, used for PCIe data communication with a PC.
- The USB UART interface is used for communication with a computer, facilitating user debugging. The serial port chip uses the Silicon Labs CP2102GM USB- UAR chip , and the USB interface uses a MINI USB interface.
- One 10/100M /1000M Ethernet RJ45 interface is used for Ethernet data exchange with computers or other network devices. The network interface chip uses the Micrel KSZ9031 industrial-grade GPHY chip.
- It has 3 standard FMC expansion ports, including 2 LPC expansion ports and 1 HPC expansion port. It can connect to various Xilinx or Alinx FMC modules (HDMI input/output modules, binocular camera modules, high-speed AD modules, etc.).
- One Micro SD card slot is used for FPGA to read, write and store data from SD cards .
- Two SMA external interfaces, with pins for connecting standard clock signals, used for external input/output signals.
- An LM75 temperature sensor chip is onboard to detect the temperature of the environment surrounding the board.
- An onboard EEPROM is used for IIC bus communication and to store some customer-defined information.
- The 10-pin 2.54mm standard JTAG port is used for downloading and debugging FPGA programs. Users can debug and download programs to the FPGA through the Xilinx programmer.
- Two 156.25MHz differential crystal oscillators are onboard to provide a reference clock for the transceiver.
- Seven light-emitting diodes (LEDs), one power indicator, four user indicator lights, and a pair of panel indicator lights.
- It has two user buttons, one reset button, and one general-purpose I/O connected to the FPGA.

01 ACKU060 Core Board

1.1 Introduction

The ACKU060 (core board model, same below) core board uses an FPGA chip based on Xilinx's Kintex UltraScale series XCKU060-2FFVA1156I. The core board uses four 1GB DDR4 chips , for a total capacity of 4GB. Additionally, the core board integrates two 128M bit QSPI FLASH chips for boot storage configuration and system files.

This core board features 6 board-to-board connectors, expanding to 354 I/O pins. Of these, 99 I/O pins on BANK64 and BANK65 operate at 3.3V, while the other BANK I/O pins operate at 1.8V. Additionally, the core board includes 20 pairs of high-speed transceiver GTH interfaces. This core board is an excellent choice for users requiring a large number of I/O pins. Furthermore, the I/O connections feature equal-length and differential traces between the FPGA chip and the interfaces, and the core board's dimensions of only 80*60 (mm) make it ideal for secondary development.

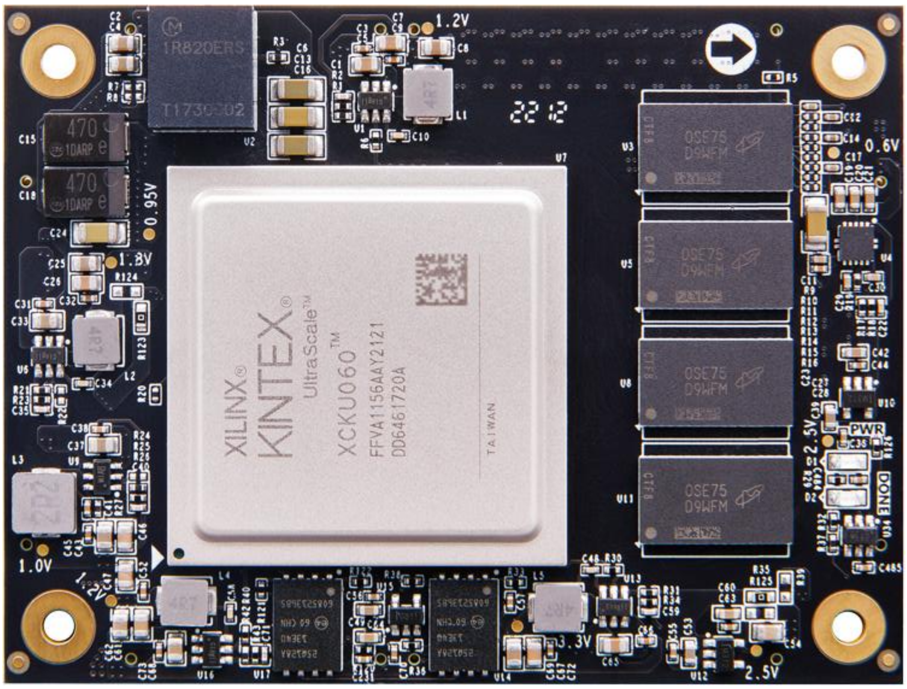

Figure 1.1.1 - Front view of ACKU060 core board

1.2 FPGA Chip

The core board uses Xilinx's Kintex UltraScale chip , model XCKU060-2FFVA1156I . It has a speed class of -2 and an industrial temperature class. This model is in an FFVA1156 package with 1156 pins and a pin pitch of 1.0mm . The naming convention for Xilinx Kintex UltraScale chips is shown in Figure 1.2.1 below .

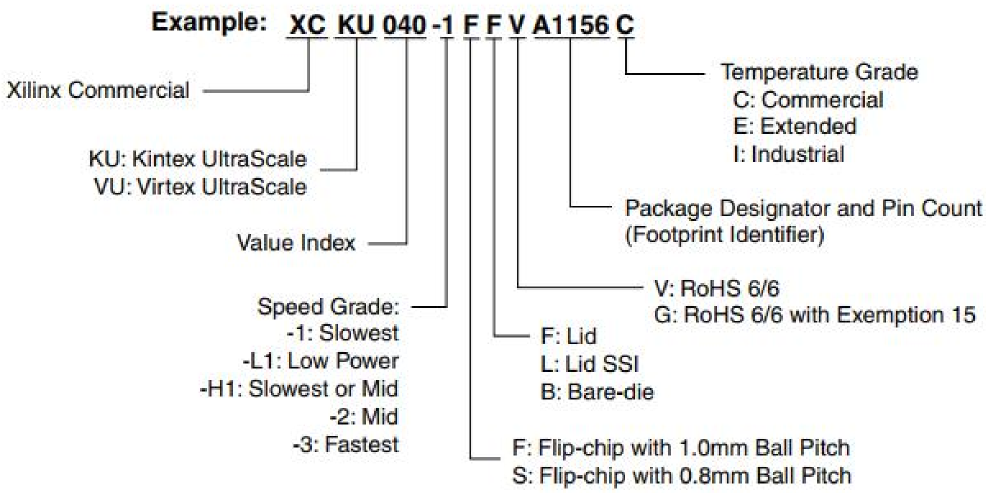

Figure 1.2.1- Kintex UltraScale FPGA Model Naming Conventions Definition

The FPGA chip XCKU060 are shown below:

Name	Specific parameters
Logic Cells	725 , 550
Lookup Tables (CLB LUTs)	331,680
Trigger (CLB flip-flops)	663,360
Block RAM (Mb) size	38.0
DSP processing unit (DSP Slices)	2,760
PCIe Gen3 x8	3
GTH Transceiver	20, 16.3Gb/s max
Speed level	-2
Temperature rating	Industrial grade

Table 1.2.1 - Main Parameters of XCKU060

1.3 DDR4 DRAM

The core board is equipped with four 1GB DDR4 chips, model NT5AD512M16C4-JRI . These four DDR4 SDRAM chips form a 64- bit bus width. Because the four DDR4 chips are connected to the FPGA, the maximum operating clock speed of the DDR4 SDRAM can reach 1200MHz. The four DDR4 memory chips are directly connected to the FPGA 's BANK 44, BANK 45, and BANK 46 interfaces. The specific configuration of the DDR4 SDRAM is shown in the table below.

Position	Chip Model	Capacity	Factory
U45, U47 , U48 , U49	NT5AD512M16C4-JRI	512 M x 16bit	Nanya

Table 1.3.1 - DDR4 SDRAM Configuration

The hardware design of DDR4 requires strict consideration of signal integrity. During circuit and PCB design, we have fully considered matching resistors/terminating resistors, trace impedance control, and trace length control to ensure the high-speed and stable operation of DDR4. The hardware connection between the FPGA and DDR4 DRAM is shown in Figure 1.3.1 :

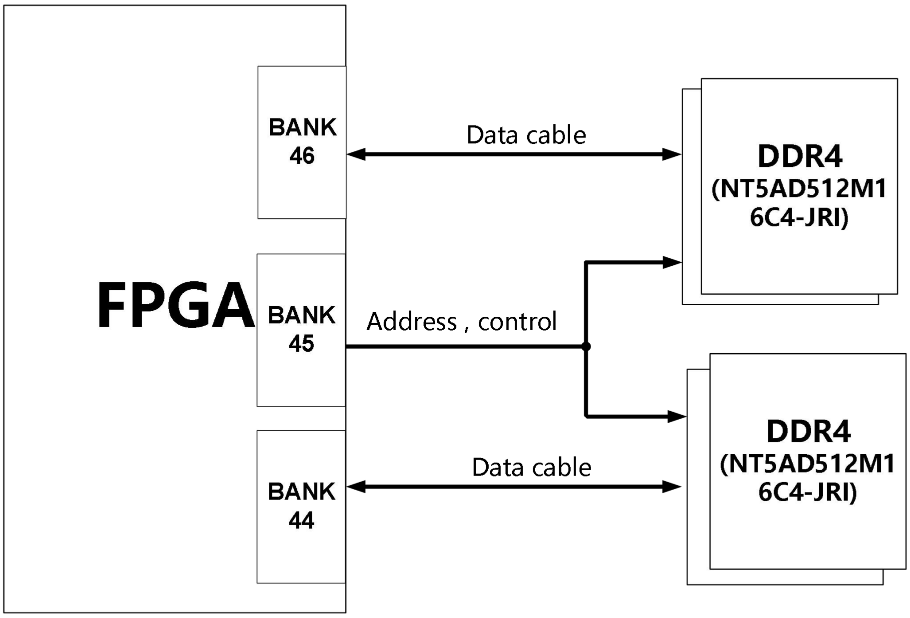

Figure 1.3.1 - Partial Schematic Diagram of DDR4 DRAM

Pin assignments for the 4 DDR4 DRAM chips:

Signal name	FPGA pin names	FPGA pin numbers
PL_DDR4_DQ0	IO_L3N_T0L_N5_AD15N_44	AE20
PL_DDR4_DQ1	IO_L2N_T0L_N3_44	AG20
PL_DDR4_DQ2	IO_L2P_T0L_N2_44	AF20
PL_DDR4_DQ3	IO_L5P_T0U_N8_AD14P_44	AE22
PL_DDR4_DQ4	IO_L3P_T0L_N4_AD15P_44	AD20
PL_DDR4_DQ5	IO_L6N_T0U_N11_AD6N_44	AG22
PL_DDR4_DQ6	IO_L6P_T0U_N10_AD6P_44	AF22

PL_DDR4_DQ7	IO_L5N_T0U_N9_AD14N_44	AE23
PL_DDR4_DQ8	IO_L8N_T1L_N3_AD5N_44	AF24
PL_DDR4_DQ9	IO_L11P_T1U_N8_GC_44	AJ23
PL_DDR4_DQ10	IO_L8P_T1L_N2_AD5P_44	AF23
PL_DDR4_DQ11	IO_L12N_T1U_N11_GC_44	AH23
PL_DDR4_DQ12	IO_L9N_T1L_N5_AD12N_44	AG25
PL_DDR4_DQ13	IO_L11N_T1U_N9_GC_44	AJ24
PL_DDR4_DQ14	IO_L9P_T1L_N4_AD12P_44	AG24
PL_DDR4_DQ15	IO_L12P_T1U_N10_GC_44	AH22
PL_DDR4_DQ16	IO_L14P_T2L_N2_GC_44	AK22
PL_DDR4_DQ17	IO_L1 7P _T2U_N 8 _AD 10P _44	AL22
PL_DDR4_DQ18	IO_L15N_T2L_N5_AD11N_44	AM20
PL_DDR4_DQ19	IO_L1 7N _T2U_N 9 _AD 10N _44	AL23
PL_DDR4_DQ20	IO_L14N_T2L_N3_GC_44	AK23
PL_DDR4_DQ21	IO_L18N_T2U_N11_AD2N_44	AL25
PL_DDR4_DQ22	IO_L15P_T2L_N4_AD11P_44	AL20
PL_DDR4_DQ23	IO_L18P_T2U_N10_AD2P_44	AL24
PL_DDR4_DQ24	IO_L20P_T3L_N2_AD1P_44	AM22
PL_DDR4_DQ25	IO_L23P_T3U_N8_44	AP24
PL_DDR4_DQ26	IO_L20N_T3L_N3_AD1N_44	AN22
PL_DDR4_DQ27	IO_L21N_T3L_N5_AD8N_44	AN24
PL_DDR4_DQ28	IO_L24P_T3U_N10_44	AN23
PL_DDR4_DQ29	IO_L23N_T3U_N9_44	AP25
PL_DDR4_DQ30	IO_L24N_T3U_N11_44	AP23

PL_DDR4_DQ31	IO_L21P_T3L_N4_AD8P_44	AM24
PL_DDR4_DQ32	IO_L2P_T0L_N2_46	AM26
PL_DDR4_DQ33	IO_L6P_T0U_N10_AD6P_46	AJ28
PL_DDR4_DQ34	IO_L2N_T0L_N3_46	AM27
PL_DDR4_DQ35	IO_L6N_T0U_N11_AD6N_46	AK28
PL_DDR4_DQ36	IO_L5P_T0U_N8_AD14P_46	AH27
PL_DDR4_DQ37	IO_L5N_T0U_N9_AD14N_46	AH28
PL_DDR4_DQ38	IO_L3P_T0L_N4_AD15P_46	AK26
PL_DDR4_DQ39	IO_L3N_T0L_N5_AD15N_46	AK27
PL_DDR4_DQ40	IO_L9N_T1L_N5_AD12N_46	AN28
PL_DDR4_DQ41	IO_L12N_T1U_N11_GC_46	AM30
PL_DDR4_DQ42	IO_L8P_T1L_N2_AD5P_46	AP28
PL_DDR4_DQ43	IO_L11N_T1U_N9_GC_46	AM29
PL_DDR4_DQ44	IO_L9P_T1L_N4_AD12P_46	AN27
PL_DDR4_DQ45	IO_L 12P _T 1 U_N 10 _ GC _4 6	AL30
PL_DDR4_DQ46	IO_L11P_T1U_N8_GC_46	AL29
PL_DDR4_DQ47	IO_L8N_T1L_N3_AD5N_46	AP29
PL_DDR4_DQ48	IO_L14P_T2L_N2_GC_46	AK31
PL_DDR4_DQ49	IO_L18P_T2U_N10_AD2P_46	AH34
PL_DDR4_DQ50	IO_L 14N _T 2L _N 3 _ GC _4 6	AK32
PL_DDR4_DQ51	IO_L15N_T2L_N5_AD11N_46	AJ31
PL_DDR4_DQ52	IO_L15P_T2L_N4_AD11P_46	AJ30
PL_DDR4_DQ53	IO_L17P_T2U_N8_AD10P_46	AH31
PL_DDR4_DQ54	IO_L18N_T2U_N11_AD2N_46	AJ34

PL_DDR4_DQ55	IO_L17N_T2U_N9_AD10N_46	AH32
PL_DDR4_DQ56	IO_L21P_T3L_N4_AD8P_46	AN31
PL_DDR4_DQ57	IO_L24P_T3U_N10_46	AL34
PL_DDR4_DQ58	IO_L23N_T3U_N9_46	AN32
PL_DDR4_DQ59	IO_L20P_T3L_N2_AD1P_46	AN33
PL_DDR4_DQ60	IO_L23P_T3U_N8_46	AM32
PL_DDR4_DQ61	IO_L24N_T3U_N11_46	AM34
PL_DDR4_DQ62	IO_L21N_T3L_N5_AD8N_46	AP31
PL_DDR4_DQ63	IO_L20N_T3L_N3_AD1N_46	AP33
PL_DDR4_DM0	IO_L1P_T0L_N0_DBC_44	AD21
PL_DDR4_DM1	IO_L7P_T1L_N0_QBC_AD13P_44	AE25
PL_DDR4_DM2	IO_L13P_T2L_N0_GC_QBC_44	AJ21
PL_DDR4_DM3	IO_L19P_T3L_N0_DBC_AD9P_44	AM21
PL_DDR4_DM4	IO_L1P_T0L_N0_DBC_46	AH26
PL_DDR4_DM5	IO_L7P_T1L_N0_QBC_AD13P_46	AN26
PL_DDR4_DM6	IO_L 13P _T 2L _N 0 _ GC _ QBC_46	AJ29
PL_DDR4_DM7	IO_L19P_T3L_N0_DBC_AD9P_46	AL32
PL_DDR4_DQS0_P	IO_L4P_T0U_N6_DBC_AD7P_44	AG21
PL_DDR4_DQS0_N	IO_L4N_T0U_N7_DBC_AD7N_44	AH21
PL_DDR4_DQS1_P	IO_L10P_T1U_N6_QBC_AD4P_44	AH24
PL_DDR4_DQS1_N	IO_L10N_T1U_N7_QBC_AD4N_44	AJ25
PL_DDR4_DQS2_P	IO_L 16P _T 2U _N 6 _ QBC _ AD3P_44	AJ20
PL_DDR4_DQS2_N	IO_L 16N _T 2U _N 7 _ QBC _ AD3N_44	AK20
PL_DDR4_DQS3_P	IO_L22P_T3U_N6_DBC_AD0P_44	AP20

PL_DDR4_DQS3_N	IO_L22N_T3U_N7_DBC_AD0N_44	AP21
PL_DDR4_DQS4_P	IO_L4P_T0U_N6_DBC_AD7P_46	AL27
PL_DDR4_DQS4_N	IO_L4N_T0U_N7_DBC_AD7N_46	AL28
PL_DDR4_DQS5_P	IO_L10P_T1U_N6_QBC_AD4P_46	AN29
PL_DDR4_DQS5_N	IO_L10N_T1U_N7_QBC_AD4N_46	AP30
PL_DDR4_DQS6_P	IO_L16P_T2U_N6_QBC_AD3P_46	AH33
PL_DDR4_DQS6_N	IO_L16N_T2U_N7_QBC_AD3N_46	AJ33
PL_DDR4_DQS7_P	IO_L22P_T3U_N6_DBC_AD0P_46	AN34
PL_DDR4_DQS7_N	IO_L22N_T3U_N7_DBC_AD0N_46	AP34
PL_DDR4_A0	IO_L18N_T2U_N11_AD2N_45	AG14
PL_DDR4_A1	IO_L23N_T3U_N9_45	AF17
PL_DDR4_A2	IO_L20P_T3L_N2_AD1P_45	AF15
PL_DDR4_A3	IO_L 16N _T 2U _N 7 _ QBC _ AD3N_45	AJ14
PL_DDR4_A4	IO_L19N_T3L_N1_DBC_AD9N_45	AD18
PL_DDR4_A5	IO_L 15P _T 2L _N 4 _ AD11P_45	AG17
PL_DDR4_A6	IO_L23P_T3U_N8_45	AE17
PL_DDR4_A7	IO_L11N_T1U_N9_GC_45	AK18
PL_DDR4_A8	IO_L24P_T3U_N10_45	AD16
PL_DDR4_A9	IO_L13P_T2L_N0_GC_QBC_45	AH18
PL_DDR4_A10	IO_L19P_T3L_N0_DBC_AD9P_45	AD19
PL_DDR4_A11	IO_L24N_T3U_N11_45	AD15
PL_DDR4_A12	IO_L 14P _T 2L _N 2 _ GC_45	AH16
PL_DDR4_A13	IO_L10N_T1U_N7_QBC_AD4N_45	AL17
PL_DDR4_BA0	IO_L18P_T2U_N10_AD2P_45	AG15

PL_DDR4_BA1	IO_L10P_T1U_N6_QBC_AD4P_45	AL18
PL_DDR4_BG0	IO_L16P_T2U_N6_QBC_AD3P_45	AJ15
PL_DDR4_WE_B	IO_L9N_T1L_N5_AD12N_45	AL15
PL_DDR4_RAS_B	IO_L8N_T1L_N3_AD5N_45	AM19
PL_DDR4_CAS_B	IO_L8P_T1L_N2_AD5P_45	AL19
PL_DDR4_CKE	IO_L14N_T2L_N3_GC_45	AJ16
PL_DDR4_ACT_B	IO_L21N_T3L_N5_AD8N_45	AF18
PL_DDR4_CLK_N	IO_L22N_T3U_N7_DBC_AD0N_45	AE15
PL_DDR4_CLK_P	IO_L22P_T3U_N6_DBC_AD0P_45	AE16
PL_DDR4_CS_B	IO_L21P_T3L_N4_AD8P_45	AE18
PL_DDR4_OTD	IO_L17P_T2U_N8_AD10P_45	AG19
PL_DDR4_PAR	IO_L20N_T3L_N3_AD1N_45	AF14
PL_DDR4_RST	IO_L 15N _T 2L _N 5 _ AD11N_45	AG16

Table 1.3.1 - DDR4 DRAM Pin Assignment

1.4 QSPI Flash

The development board comes with two 128 Mb Quad - SPI FLASH chips, model N25Q128A , which use the 3.3V CMOS voltage standard. Due to the non-volatile nature of QSPI FLASH, it can store the FPGA configuration binary file and other user data files. Specific models and parameters of the QSPI FLASH are shown in Table 1.4.1 .

Position	Chip type	capacity	factory
U14	N25Q128A	128M bit	Numonyx

Table 1.4.1 - QSPI Flash Models and Parameters

The QSPI FLASH is connected to a dedicated pin on BANK 0 of the FPGA chip. The clock pin is connected to CCLK0 of BANK 0, and the other data signals are connected to D00~D03 and FCS pins of BANK 0, respectively. Figure 1.4.2 shows a schematic diagram of the connection between the QSPI Flash and the FPGA chip.

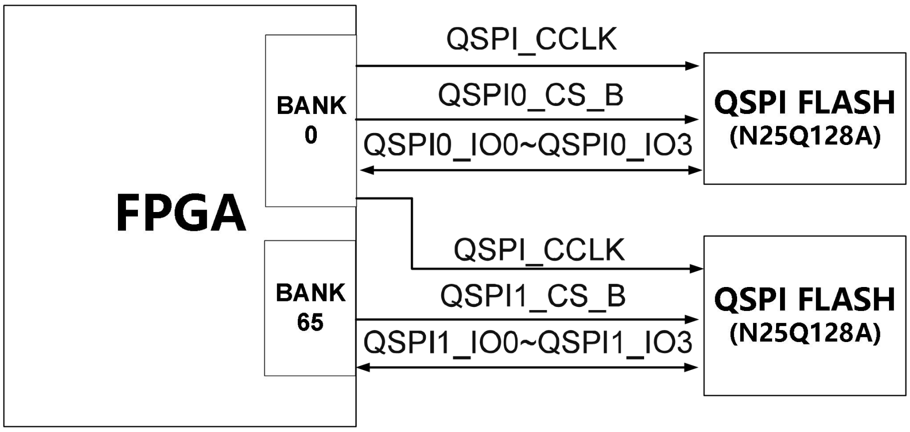

Figure 1.4.2 - QSPI Flash Connection Diagram

Chip pin assignment:

Signal name	FPGA pin names	FPGA pin numbers
QSPI_CCLK	CCLK_0	AA9
QSPI0_CS_B	RDWR_FCS_B_0	U7
QSPI0_IO0	D00_MOSI_0	AC7
QSPI0_IO1	D01_DIN_0	AB7
QSPI0_IO2	D02_0	AA7
QSPI0_IO3	D03_0	Y7
QSPI 1 _CS_B	IO_L2N_T0L_N3_FWE_FCS2_B_65	G26
QSPI 1 _IO0	IO_L22P_T3U_N6_DBC_AD0P_D04_65	M20
QSPI 1 _IO1	IO_L22N_T3U_N7_DBC_AD0N_D05_65	L20
QSPI 1 _IO2	IO_L21P_T3L_N4_AD8P_D06_65	R21
QSPI 1 _IO3	IO_L21N_T3L_N5_AD8N_D07_65	R22

Table 1.4.1 - QSPI Flash Pin Assignment

1.5 Clock Configuration

200MHz differential clock source

The core board provides a differential 200MHz clock source to supply the system clock for the FPGA . The differential output of the crystal oscillator is connected to the FPGA BANK45, and this clock can be used to drive the operating clock of the FPGA's internal DDR controller and other user logic circuits.

System clock pin assignment table:

Signal name	FPGA pins
PL_CLK0_P	AK17
PL_CLK0_N	AK16

Table 1.5.1 - System Clock Pin Assignment

1.6 LED Lights

The core board has two red LEDs: one is the power indicator (PWR), and the other is the configuration LED (DONE). Both the power indicator and the DONE LED light up upon power-on; the DONE LED turns off after the FPGA configuration process is complete. A schematic diagram of the LED hardware connection is shown in the figure.

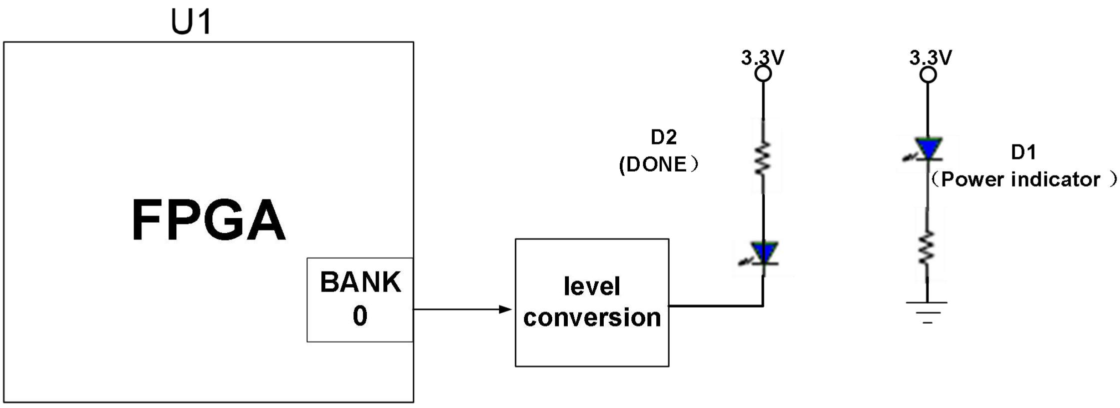

Figure 1.6.1 - Hardware connection diagram of LED lights on the core board

1.7 Power Supply

The ACKU060 core board is powered by DC 12V and is connected to the baseboard. The power supply design diagram on the board is shown in Figure 1.7.1 below :

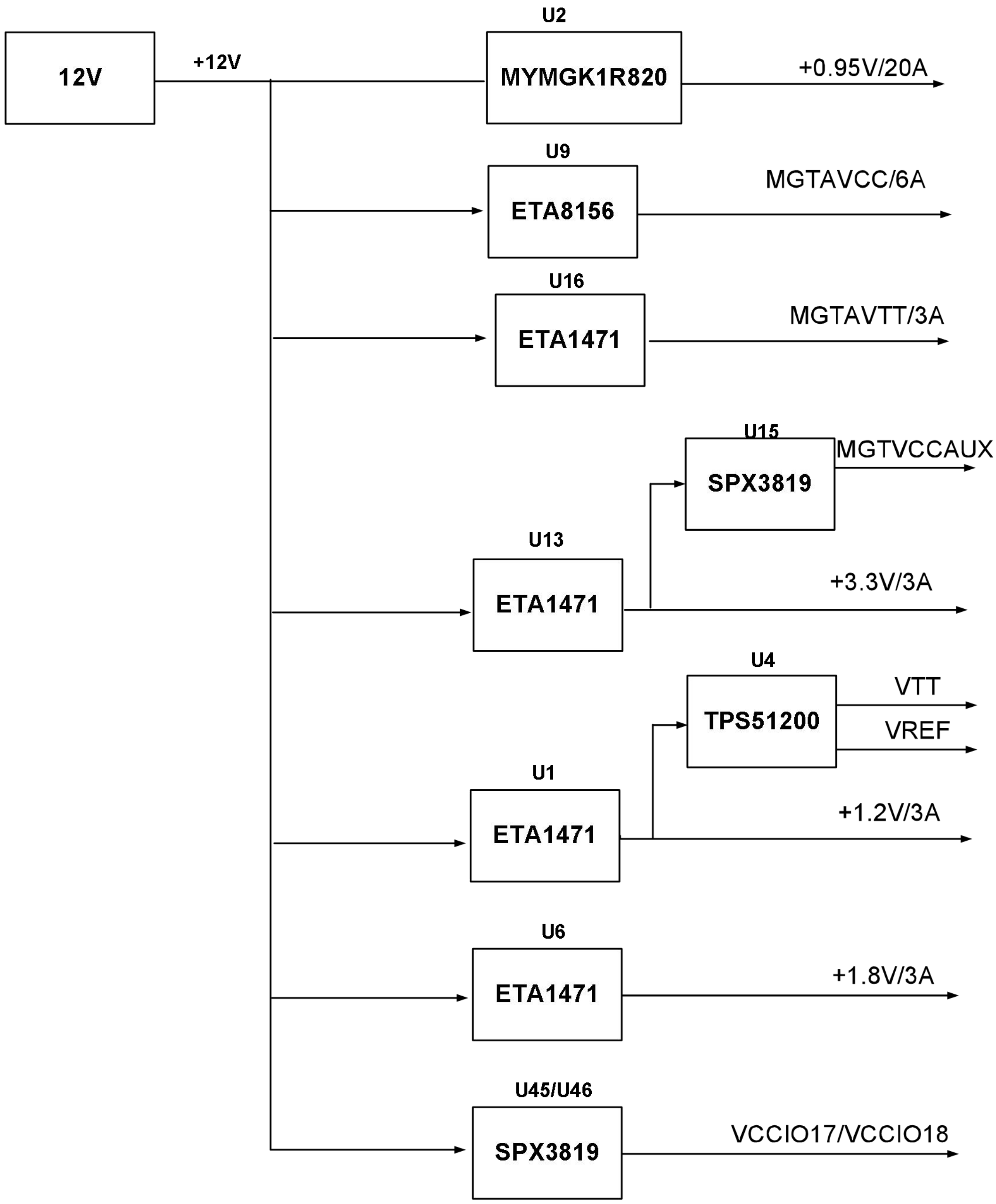

Figure 1.7.1 - Power interface section in the schematic diagram

Power supply generates a +0.95V FPGA core power supply via a MYMGK1R820ERSR DC-DC power chip, providing up to 20A of current, far exceeding the core voltage's current requirements. This +12V power supply is then used by an ETA1471 DC-DC chip to generate four power supplies: +1.2V, +1.8V , +3.3V, and MGTAVTT. The MGTAVCC power supply used by the GTX transceiver is generated by an ETA8156 DC-DC chip, and an auxiliary +1.8V power supply for the GTX is generated by an SPX3819-1-8 LDO chip . The VTT and VREF voltages for the DDR4 are generated by a TPS51200.

1.8 Structure Diagram

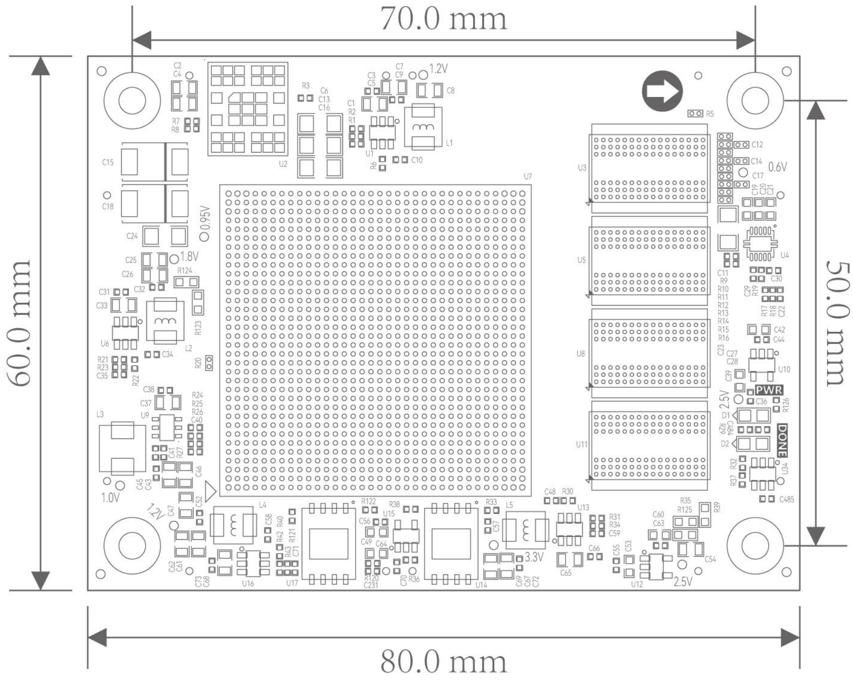

Table 1.8.1 - Top View

1.9 Connector Pin Definitions

The core board has a total of 6 high-speed expansion ports, using 4 120-pin connectors (J1, J3, J4, J5) and 2 80-pin connectors (J2, J6) to connect to the baseboard. The connectors are Panasonic AXK5A2137YG and AXK580137YG ,corresponding to the baseboard's connector models AXK6A2337YG and AXK680337YG . The J1 connector is for BANK66 and BANK68 signals, with a standard voltage level of 1.8V.

J1 connector pin assignment

J1 pin	Signal name	pin number	J1 pin	Signal name	pin number
1	B66_L3_N	C8	2	B66_L1_N	E8
3	B66_L3_P	D8	4	B66_L1_P	F8
5	B66_L7_N	K8	6	B66_L2_N	A9
7	B66_L7_P	L8	8	B66_L2_P	B9
9	GND	-	10	GND	-

11	B66_L9_N	H8	12	B66_L4_N	A10
13	B66_L9_P	J8	14	B66_L4_P	B10
15	B66_L8_N	H9	16	B66_L11_N	F9
17	B66_L8_P	J9	18	B66_L11_P	G9
19	GND	-	20	GND	-
21	B66_L10_N	J10	22	B66_L12_N	F10
23	B66_L10_P	K10	24	B66_L12_P	G10
25	B66_L5_N	C9	26	B66_L6_N	D10
27	B66_L5_P	D9	28	B66_L6_P	E10
29	GND	-	30	GND	-
31	B66_L17_N	K12	32	B66_L13_N	G11
33	B66_L17_P	L12	34	B66_L13_P	H11
35	B66_L19_N	D11	36	B66_L15_N	J11
37	B66_L19_P	E11	38	B66_L15_P	K11
39	GND	-	40	GND	-
41	B66_L16_N	K13	42	B66_L14_N	G12
43	B66_L16_P	L13	44	B66_L14_P	H12
45	B66_L20_N	B12	46	B66_L18_N	H13
47	B66_L20_P	C12	48	B66_L18_P	J13
49	GND	-	50	GND	-
51	B66_L22_N	E13	52	B66_L21_N	B11
53	B66_L22_P	F13	54	B66_L21_P	C11
55	B66_L24_N	C13	56	B66_L23_N	A12
57	B66_L24_P	D13	58	B66_L23_P	A13

59	GND	-	60	GND	-
61	B68_L9_N	F14	62	B68_L19_N	J14
63	B68_L9_P	F15	64	B68_L19_P	J15
65	B68_L8_N	D15	66	B68_L21_N	K15
67	B68_L8_P	E15	68	B68_L21_P	L15
69	GND	-	70	GND	-
71	B68_L15_N	G14	72	B68_L11_N	D16
73	B68_L15_P	G15	74	B68_L11_P	E16
75	B68_L20_N	K17	76	B68_L23_N	J16
77	B68_L20_P	K18	78	B68_L23_P	K16
79	GND	-	80	GND	-
81	B68_L16_N	F19	82	B68_L10_N	D18
83	B68_L16_P	G19	84	B68_L10_P	D19
85	B68_L18_N	H18	86	B68_L1_N	A14
87	B68_L18_P	H19	88	B68_L1_P	B14
89	GND	-	90	GND	-
91	B68_L22_N	J18	92	B68_L3_N	A15
93	B68_L22_P	J19	94	B68_L3_P	B15
95	B68_L24_N	L18	96	B68_L5_N	B16
97	B68_L24_P	L19	98	B68_L5_P	B17
99	GND	-	100	GND	-
101	B68_L13_N	G16	102	B68_L7_N	C14
103	B68_L13_P	G17	104	B68_L7_P	D14
105	B68_L14_N	F17	106	B68_L6_N	C17

107	B68_L14_P	F18	108	B68_L6_P	C18
109	GND	-	110	GND	-
111	B68_L12_N	E17	112	B68_L2_N	A18
113	B68_L12_P	E18	114	B68_L2_P	A19
115	B68_L17_N	H16	116	B68_L4_N	B19
117	B68_L17_P	H17	118	B68_L4_P	C19
119	GND	-	120	GND	-

Table 1.9.1 - J1 Connector Pin Assignment

The J2 connector has 80 pins and connects to the high-speed differential signal of the transceiver BANK226~228.

J2 connector pin assignment

J2 pin	Signal name	pin number	J2 pin	Signal name	pin number
1	GND	-	2	GND	-
3	226_TX2_N	U3	4	226_RX2_N	T1
5	226_TX2_P	U4	6	226_RX2_P	T2
7	GND	-	8	GND	-
9	226_TX3_N	R3	10	226_RX3_N	P1
11	226_TX3_P	R4	12	226_RX3_P	P2
13	GND	-	14	GND	-
15	226_CLK1_N	T5	16	226_CLK0_N	V5
17	226_CLK1_P	T6	18	226_CLK0_P	V6
19	GND	-	20	GND	-
21	227_TX0_P	N4	22	227_RX0_P	M2
23	227_TX0_N	N3	24	227_RX0_N	M1
25	GND	-	26	GND	-

27	227_TX1_P	L4	28	227_RX1_P	K2
29	227_TX1_N	L3	30	227_RX1_N	K1
31	GND	-	32	GND	-
33	227_TX2_P	J4	34	227_RX2_P	H2
35	227_TX2_N	J3	36	227_RX2_N	H1
37	GND	-	38	GND	-
39	227_TX3_P	G4	40	227_RX3_P	F2
41	227_TX3_N	G3	42	227_RX3_N	F1
43	GND	-	44	GND	-
45	227_CLK1_P	M6	46	227_CLK0_P	P6
47	227_CLK1_N	M5	48	227_CLK0_N	P5
49	GND	-	50	GND	-
51	228_TX0_P	F6	52	228_RX0_P	E4
53	228_TX0_N	F5	54	228_RX0_N	E3
55	GND	-	56	GND	-
57	228_TX1_P	D6	58	228_RX1_P	D2
59	228_TX1_N	D5	60	228_RX1_N	D1
61	GND	-	62	GND	-
63	228_TX2_P	C4	64	228_RX2_P	B2
65	228_TX2_N	C3	66	228_RX2_N	B1
67	GND	-	68	GND	-
69	228_TX3_P	B6	70	228_RX3_P	A4
71	228_TX3_N	B5	72	228_RX3_N	A3
73	GND	-	74	GND	-

75	228_CLK1_P	H6	76	228_CLK0_P	K6
77	228_CLK1_N	H5	78	228_CLK0_N	K5
79	GND	-	80	GND	-

Table 1.9.2 - J2 Connector Pin Assignment

J3 represents the high-speed differential signal from transceivers BANK224~226 and part of the signals from BANK64 and BANK65.

J3 connector pin assignment

J3 pin	Signal name	pin number	J3 pin	Signal name	pin number
1	B64_L7_N	AF13	2	B64_L21_N	AL9
3	B64_L7_P	AE13	4	B64_L21_P	AK10
5	B64_L11_N	AH12	6	B64_L24_N	AL8
7	B64_L11_P	AG12	8	B64_L24_P	AK8
9	GND	L7	10	GND	-
11	B64_L9_N	AF12	12	B64_L12_N	AH11
13	B64_L9_P	AE12	14	B64_L12_P	AG11
15	B64_L13_N	AG10	16	B64_L14_N	AG9
17	B64_L13_P	AF10	18	B64_L14_P	AF9
19	GND	L7	20	GND	-
21	B64_L10_N	AE11	22	B64_L15_N	AF8
23	B64_L10_P	AD11	24	B64_L15_P	AE8
25	B64_L18_N	AH8	26	B64_L16_N	AE10
27	B64_L18_P	AH9	28	B64_L16_P	AD10
29	GND	L7	30	GND	-
31	B64_L17_N	AD8	32	FPGA_TCK	AC9
33	B64_L17_P	AD9	34	FPGA_TDO	U9

35	B64_L23_N	AJ8	36	FPGA_TMS	W9
37	B64_L23_P	AJ9	38	FPGA_TDI	V9
39	GND	L7	40	GND	-
41	B65_T0U	H23	42	B66_T3U	E12
43	B65_T3U	K22	44	B66_T2U	F12
45	B65_T1U	N23	46	B66_T1U	L9
47	B65_T2U	N27	48	NC	-
49	GND	L7	50	GND	-
51	224_TX0_N	AN3	52	224_RX0_N	AP1
53	224_TX0_P	AN4	54	224_RX0_P	AP2
55	GND	L7	56	GND	-
57	224_TX1_N	AM5	58	224_RX1_N	AM1
59	224_TX1_P	AM6	60	224_RX1_P	AM2
61	GND	L7	62	GND	-
63	224_TX2_N	AL3	64	224_RX2_N	AK1
65	224_TX2_P	AL4	66	224_RX2_P	AK2
67	GND	L7	68	GND	-
69	224_TX3_N	AK5	70	224_RX3_N	AJ3
71	224_TX3_P	AK6	72	224_RX3_P	AJ4
73	GND	L7	74	GND	-
75	224_CLK1_N	AD5	76	224_CLK0_N	AF5
77	224_CLK1_P	AD6	78	224_CLK0_P	AF6
79	GND	L7	80	GND	-
81	225_TX0_N	AH5	82	225_RX0_N	AH1

83	225_TX0_P	AH6	84	225_RX0_P	AH2
85	GND	L7	86	GND	-
87	225_TX1_N	AG3	88	225_RX1_N	AF1
89	225_TX1_P	AG4	90	225_RX1_P	AF2
91	GND	L7	92	GND	-
93	225_TX2_N	AE3	94	225_RX2_N	AD1
95	225_TX2_P	AE4	96	225_RX2_P	AD2
97	GND	L7	98	GND	-
99	225_TX3_N	AC3	100	225_RX3_N	AB1
101	225_TX3_P	AC4	102	225_RX3_P	AB2
103	GND	L7	104	GND	-
105	225_CLK1_N	Y5	106	225_CLK0_N	AB5
107	225_CLK1_P	Y6	108	225_CLK0_P	AB6
109	GND	L7	110	GND	-
111	226_TX0_N	AA3	112	226_RX0_N	Y1
113	226_TX0_P	AA4	114	226_RX0_P	Y2
115	GND	L7	116	GND	-
117	226_TX1_N	W3	118	226_RX1_N	V1
119	226_TX1_P	W4	120	226_RX1_P	V2

Table 1.9.3 - J3 Connector Pin Assignment

J4 connects the signals of BANK48 and part of BANK64.

J4 connector pin assignment

J4 pin	Signal name	pin number	J4 pin	Signal name	pin number
1	B48_L8_N	AG34	2	B48_T2U	AA33
3	B48_L8_P	AF33	4	B48_T1U	AE31

5	B48_L7_N	AG32	6	B48_T3U	V32
7	B48_L7_P	AG31	8	B47_T3U	U29
9	GND	-	10	GND	-
11	B48_L10_N	AF34	12	B48_L18_N	AD33
13	B48_L10_P	AE33	14	B48_L18_P	AC33
15	B48_L9_N	AF32	16	B48_L23_N	V34
17	B48_L9_P	AE32	18	B48_L23_P	U34
19	GND	-	20	GND	-
21	B48_L12_N	AC32	22	B48_L21_N	W34
23	B48_L12_P	AC31	24	B48_L21_P	V33
25	B48_L11_N	AD31	26	B48_L17_N	AB34
27	B48_L11_P	AD30	28	B48_L17_P	AA34
29	GND	-	30	GND	-
31	B48_L13_N	AB32	32	B48_L15_N	AD34
33	B48_L13_P	AA32	34	B48_L15_P	AC34
35	B48_L4_N	AG29	36	B48_L19_N	Y33
37	B48_L4_P	AF29	38	B48_L19_P	W33
39	GND	-	40	GND	-
41	B48_L2_N	AF28	42	B48_L6_N	AG30
43	B48_L2_P	AE28	44	B48_L6_P	AF30
45	B48_L1_N	AF27	46	B48_L5_N	AE30
47	B48_L1_P	AE27	48	B48_L5_P	AD29
49	GND	-	50	GND	-
51	B48_L3_N	AD28	52	B48_L16_N	AB29

53	B48_L3_P	AC28	54	B48_L16_P	AA29
55	B48_L14_N	AB31	56	B48_L24_N	W31
57	B48_L14_P	AB30	58	B48_L24_P	V31
59	GND	-	60	GND	-
61	B48_L20_N	Y30	62	NC	-
63	B48_L20_P	W30	64	NC	-
65	B48_L22_N	Y32	66	NC	-
67	B48_L22_P	Y31	68	NC	-
69	GND	-	70	GND	-
71	B47_T1U	Y22	72	FPGA_CCLK	AA9
73	B47_T2U	Y21	74	FPGA_DIN	AB7
75	NC	-	76	FPGA_INIT_B	V7
77	NC	-	78	FPGA_PROGRAM_B	T7
79	GND	L7	80	GND	-
81	NC	-	82	FPGA_DONE	N7
83	NC	-	84	NC	-
85	NC	-	86	NC	-
87	NC	-	88	POWER_PG	-
89	GND	-	90	GND	-
91	B64_L8_N	AJ13	92	B64_T1U	AJ11
93	B64_L8_P	AH13	94	B64_T3U	AM9
95	B64_L6_N	AL13	96	B64_T0U	AK11
97	B64_L6_P	AK13	98	B64_T2U	AJ10
99	GND	-	100	GND	-

101	B64_L1_N	AP10	102	B64_L2_N	AP13
103	B64_L1_P	AP11	104	B64_L2_P	AN13
105	B64_L4_N	AN12	106	B64_L22_N	AP8
107	B64_L4_P	AM12	108	B64_L22_P	AN8
109	GND	-	110	GND	-
111	B64_L20_N	AP9	112	B64_L19_N	AM10
113	B64_L20_P	AN9	114	B64_L19_P	AL10
115	B64_L3_N	AN11	116	B64_L5_N	AL12
117	B64_L3_P	AM11	118	B64_L5_P	AK12
119	GND	-	120	GND	-

Table 1.9.4 - J4 Connector Pin Assignment

J5 connects the signals of BANK47 and part of BANK65.

J5 connector pin assignment

J5 pin	Signal name	pin number	J5 pin	Signal name	pin number
1	B65_L10_N	K23	2	NC	-
3	B65_L10_P	L22	4	NC	-
5	B65_L6_N	H24	6	B65_L23_N	M21
7	B65_L6_P	J23	8	B65_L23_P	N21
9	GND	L7	10	GND	-
11	B65_L19_N	M22	12	NC	-
13	B65_L19_P	N22	14	B65_L2_P	G25
15	B65_L9_N	K25	16	B65_L1_N	G27
17	B65_L9_P	L25	18	B65_L1_P	H27
19	GND	L7	20	GND	-

21	B65_L24_N	K21	22	B65_L5_N	H26
23	B65_L24_P	K20	24	B65_L5_P	J26
25	B65_L12_N	M24	26	B65_L4_N	J25
27	B65_L12_P	N24	28	B65_L4_P	J24
29	GND	L7	30	GND	-
31	B65_L20_N	P21	32	B65_L3_N	K27
33	B65_L20_P	P20	34	B65_L3_P	K26
35	B65_L7_N	L27	36	B65_L11_N	M26
37	B65_L7_P	M27	38	B65_L11_P	M25
39	GND	L7	40	GND	-
41	B65_L13_N	N26	42	B65_L18_N	P23
43	B65_L13_P	P26	44	B65_L18_P	R23
45	B65_L14_N	P25	46	B65_L15_N	R27
47	B65_L14_P	P24	48	B65_L15_P	T27
49	GND	-	50	GND	-
51	B65_L8_N	L24	52	B65_L17_N	R26
53	B65_L8_P	L23	54	B65_L17_P	R25
55	NC	-	56	B65_L16_N	T25
57	NC	-	58	B65_L16_P	T24
59	GND	L7	60	GND	-
61	B47_L11_N	AA23	62	B47_L19_N	V28
63	B47_L11_P	Y23	64	B47_L19_P	V27
65	B47_L14_N	Y25	66	B47_L22_N	U27
67	B47_L14_P	W25	68	B47_L22_P	U26

69	GND	-	70	GND	-
71	B47_L7_N	AB22	72	B47_L20_N	U25
73	B47_L7_P	AA22	74	B47_L20_P	U24
75	B47_L21_N	Y28	76	B47_L17_N	T23
77	B47_L21_P	W28	78	B47_L17_P	T22
79	GND	-	80	GND	-
81	B47_L3_N	AC24	82	B47_L15_N	U22
83	B47_L3_P	AB24	84	B47_L15_P	U21
85	B47_L23_N	W29	86	B47_L24_N	W26
87	B47_L23_P	V29	88	B47_L24_P	V26
89	GND	-	90	GND	-
91	B47_L10_N	AC21	92	B47_L13_N	W24
93	B47_L10_P	AB21	94	B47_L13_P	W23
95	B47_L5_N	AB27	96	B47_L1_N	Y27
97	B47_L5_P	AA27	98	B47_L1_P	Y26
99	GND	-	100	GND	-
101	B47_L9_N	AB20	102	B47_L12_N	AA25
103	B47_L9_P	AA20	104	B47_L12_P	AA24
105	B47_L4_N	AC27	106	B47_L6_N	AB26
107	B47_L4_P	AC26	108	B47_L6_P	AB25
109	GND	-	110	GND	-
111	B47_L8_N	AC23	112	B47_L16_N	V23
113	B47_L8_P	AC22	114	B47_L16_P	V22
115	B47_L2_N	AD26	116	B47_L18_N	W21

117	B47_L2_P	AD25	118	B47_L18_P	V21
119	GND	-	120	GND	-

Table 1.9.5 - J5 Connector Pin Assignment

J6 connects to a 12V power supply, BANK66, and some signals from BANK68.

J6 connector pin assignment

J6 pin	Signal name	pin number	J6 pin	Signal name	pin number
1	+12V	-	2	+12V	-
3	+12V	-	4	+12V	-
5	+12V	-	6	+12V	-
7	+12V	-	8	+12V	-
9	+12V	-	10	+12V	-
11	GND	-	12	GND	-
13	B67_L17_N	A20	14	B67_L8_N	A25
15	B67_L17_P	B20	16	B67_L8_P	B25
17	B67_L16_N	C22	18	B67_L6_N	A28
19	B67_L16_P	C21	20	B67_L6_P	A27
21	GND	-	22	GND	-
23	B67_L15_N	B22	24	B67_L13_N	C23
25	B67_L15_P	B21	26	B67_L13_P	D23
27	B67_L11_N	D25	28	B67_L12_N	C24
29	B67_L11_P	E25	30	B67_L12_P	D24
31	GND	-	32	GND	-
33	B67_L18_N	D21	34	B67_L4_N	A29
35	B67_L18_P	D20	36	B67_L4_P	B29

37	B67_L20_N	E21	38	B67_L2_N	B27
39	B67_L20_P	E20	40	B67_L2_P	C27
41	GND	-	42	GND	-
43	B67_L14_N	E23	44	B67_L1_N	E27
45	B67_L14_P	E22	46	B67_L1_P	F27
47	B67_L22_N	F20	48	B67_L10_N	A24
49	B67_L22_P	G20	50	B67_L10_P	B24
51	GND	-	52	GND	-
53	B67_L19_N	F25	54	B67_L9_N	B26
55	B67_L19_P	G24	56	B67_L9_P	C26
57	B67_L24_N	G21	58	B67_L5_N	C28
59	B67_L24_P	H21	60	B67_L5_P	D28
61	GND	-	62	GND	-
63	B67_L21_N	F24	64	B67_L3_N	D29
65	B67_L21_P	F23	66	B67_L3_P	E28
67	B67_L23_N	F22	68	B67_L7_N	D26
69	B67_L23_P	G22	70	B67_L7_P	E26
71	GND	-	72	GND	-
73	B68_T1U	C16	74	B67_T1U	A23
75	B68_T2U	H14	76	B67_T2U	A22
77	B68_T3U	L17	78	B67_T3U	H22
79	NC	-	80	NC	-

Table 1.9.6 - J6 Connector Pin Assignment

02 Expansion Board

2.1 Introduction

Through the preceding functional overview, we can understand the functions of the expansion board.

- 2 fiber optic interface
- 1 PCIe x8 interface
- 1 USB UART interface
- 1 Ethernet RJ45 interface
- 3 standard FMC expansion ports
- One Micro SD card slot.
- 2 SMA interface
- EEPROM and temperature sensor
- JTAG debugging interface.
- 7 light-emitting diodes (LEDs)
- Two user buttons.

2.2 PCIex8 Interface

The AXKU062 is equipped with a PCIe 3.0 x8 interface, with 8 pairs of transceivers connecting to the PCIe x8 gold fingers, enabling PCIe x8, PCIe x4, PCIe x2, PCIe x1 data communication.

The PCIe interface transmits and receives signals directly to the GTH transceiver of the FPGA BANK224 and BANK225 . The 8 TX and RX signals are connected to the FPGA transceiver in a differential signal manner, and the single-channel communication rate can reach up to 8Gbps bandwidth.

The PCIe interface of the development board is shown in the figure below , where the TX signal is connected using AC coupling mode.

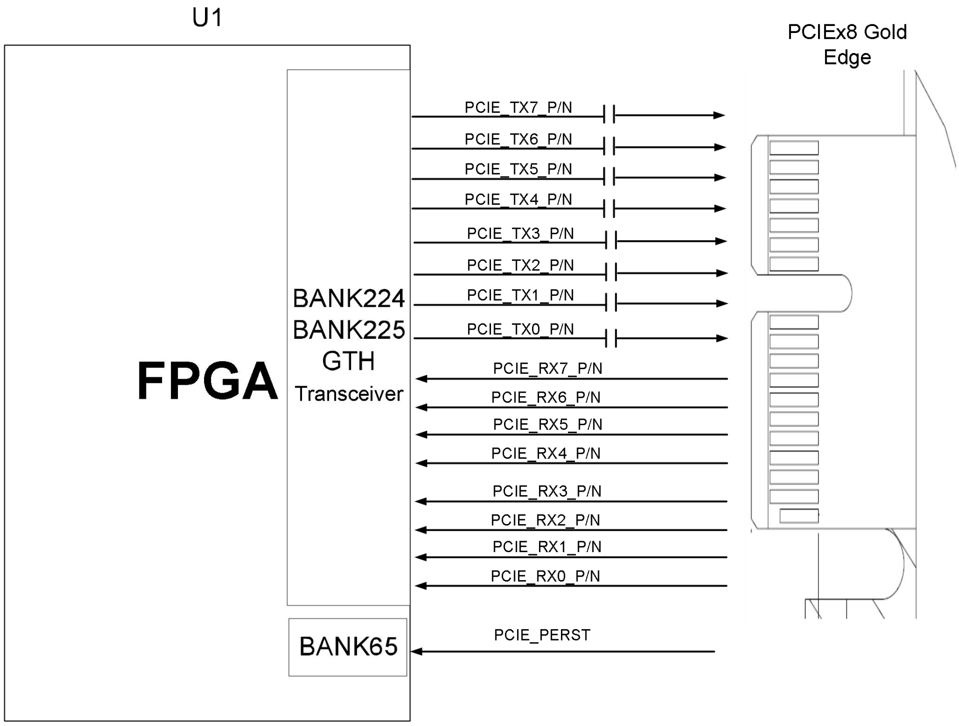

Figure 2.2.1 - Schematic diagram of PCIe slot design

The pin assignments for the PCIe x8 interface FPGA are as follows:

Signal name	FPGA pin names	pin number	Remark
PCIE_RX0_N	MGTHRXN3_225	AB1	PCIe channel 0 data reception negative
PCIE_RX0_P	MGTHRXP3_225	AB2	PCIe channel 0 data reception is positive.
PCIE_RX1_N	MGTHRXN2_225	AD1	PCIe Channel 1 data reception is negative
PCIE_RX1_P	MGTHRXP2_225	AD2	PCIe Channel 1 data reception is positive.
PCIE_RX2_N	MGTHRXN1_225	AF1	PCIe lane 2 data reception negative
PCIE_RX2_P	MGTHRXP1_225	AF2	PCIe Channel 2 data reception is positive
PCIE_RX3_N	MGTHRXN0_225	AH1	PCIe lane 3 data reception negative
PCIE_RX3_P	MGTHRXP0_225	AH2	PCIe channel 3 data reception is positive
PCIe RX4 N	MGTHRXN3_224	AJ3	PCIe lane 4 data reception negative
PCIe RX4 P	MGTHRXP3_224	AJ4	PCIe channel 4 data reception is positive
PCIE_RX5_N	MGTHRXN2_224	AK1	PCIe channel 5 data reception negative
PCIe RX5 P	MGTHRXP2_224	AK2	PCIe channel 5 data reception is positive.

PCIE_RX6_N	MGTHRXN1_224	AM1	PCIe channel 6 data reception negative
PCIE_RX6_P	MGTHRXP1_224	AM2	PCIe channel 6 data reception is positive.
PCIE_RX7_N	MGTHRXN0_224	AP1	PCIe lane 7 data reception negative
PCIe RX7 P	MGTHRXP0_224	AP2	PCIe channel 7 data reception is positive.
PCIE_TX0_N	MGTHTXN3_225	AC3	PCIe channel 0 data transmission negative
PCIE_TX0_P	MGTHTXP3_225	AC4	PCIe channel 0 data transmission is positive.
PCIE_TX1_N	MGTHTXN2_225	AE3	PCIe Channel 1 Data Transmission Negative
PCIE_TX1_P	MGTHTXP2_225	AE4	PCIe Channel 1 data transmission is positive.
PCIE_TX2_N	MGTHTXN1_225	AG3	PCIe Channel 2 Data Transmission Negative
PCIE_TX2_P	MGTHTXP1_225	AG4	PCIe Channel 2 data transmission is positive
PCIE_TX3_N	MGTHTXN0_225	AH5	PCIe Channel 3 Data Transmission Negative
PCIE_TX3_P	MGTHTXP0_225	AH6	PCIe Channel 3 data transmission is positive
PCIE_TX4_N	MGTHTXN3_224	AK5	PCIe Channel 4 Data Transmission Negative
PCIe TX4 P	MGTHTXP3_224	AK6	PCIe Channel 4 data transmission is positive.
PCIE_TX5_N	MGTHTXN2_224	AL3	PCIe channel 5 data transmission negative
PCIe TX5 P	MGTHTXP2_224	AL4	PCIe channel 5 data transmission is positive
PCIE_TX6_N	MGTHTXN1_224	AM5	PCIe channel 6 data transmission negative
PCIE_TX6_P	MGTHTXP1_224	AM6	PCIe channel 6 data transmission is positive
PCIE_TX7_N	MGTHTXN0_224	AN3	PCIe channel 7 data transmission negative
PCIE_TX7_P	MGTHTXP0_224	AN4	PCIe channel 7 data transmission is positive.
PCIE_CLK_N	MGTREFCLK0N_225	AB5	PCIe channel reference clock negative
PCIE_CLK_P	MGTREFCLK0P_225	AB6	PCIe channel reference clock is positive
PCIE_PERST	IO_T3U_N12_PERST N0_65	K22	PCIe card reset signal

Table 2.2.1 - PCIe x8 Pin Assignment

2.3 SFP+ Fiber Optic Interface

The AXKU062 development board has two SFP fiber optic interfaces. Users can purchase SFP optical modules (1.25G, 2.5G, and 10G optical modules are available on the market) and insert them into these two fiber optic interfaces for fiber optic data communication. The two fiber optic interfaces are connected to the two RX/TX channels of the BANK226 GTH transceiver on the FPGA. Both TX and RX signals are differentially connected between the FPGA and the optical module, with each TX transmit and RX receive data rate reaching up to 16.3Gb/s. The reference clock for the BANK226 GTH transceiver is provided by a 156.25MHz differential crystal oscillator. A schematic diagram of the FPGA and SFP fiber optic design is shown in Figure 2.3.1 below .

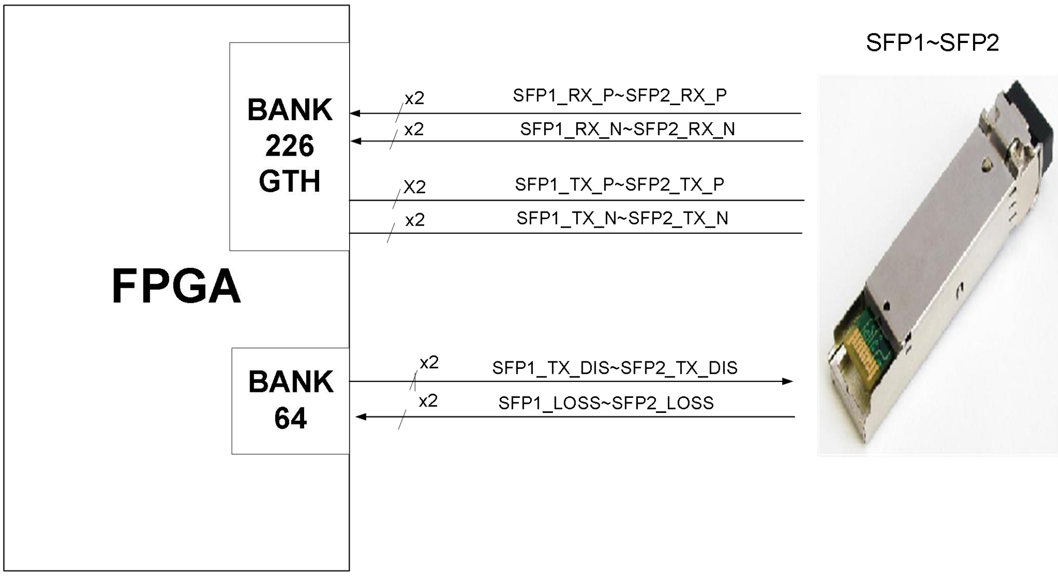

Figure 2.3.1 - Schematic diagram of fiber optic design

The FPGA pin assignments for the first fiber optic interface are as follows:

Network Name	FPGA pins	Remark
SFP 1 _TX_P	U4	SFP optical module data transmission Positive
SFP 1 _TX_ N	U3	SFP optical module data transmission Negative
SFP 1 _ R X_P	T2	SFP optical module data receive Positive
SFP 1 _ R X _ N	T1	SFP optical module data receive Negative
SFP 1 _ TX_DIS	AN11	SFP optical module transmit disable, active high
SFP 1 _ LOSS	AP9	SFP receive LOSS signal, high indicates no optical signal received

Table 2.3.1 - FPGA Pin Assignment for the First Fiber Optic Interface

The FPGA pin assignments for the second fiber optic interface are as follows:

Network Name	FPGA pins	Remark
SFP 2 _TX_P	W4	SFP optical module data transmit Positive
SFP 2 _TX_ N	W3	SFP optical module data transmit Negative
SFP 2 _ R X_P	V2	SFP optical module data receive Positive
SFP 2 _ RX _ N	V1	SFP optical module data receive Negative
SFP 2 _ TX_DIS	AM11	SFP optical module transmit disable, active high
SFP 2 _ LOSS	AN9	SFP receive LOSS signal, high indicates no optical signal received

Table 2.3.2 - FPGA Pin Assignment for the Second Fiber Optic Interface

2.4 Gigabit Ethernet Interface

The AXKU062 has one Gigabit Ethernet interface, and the GPHY chip uses the Micrel KSZ9031RNX Ethernet chip to provide network communication services for users. The KSZ9031RNX chip supports network transmission rates of 10/100/1000 Mbps and communicates with the system 's MAC layer through the RGMII interface . The KSZ9031RNX supports MDI/MDX auto-adaptation, various speed auto-adaptation, Master/ Slave auto-adaptation, and supports MDIO bus for PHY register management.

Upon power-up, the KSZ9031RNX detects the voltage levels of specific I/O pins to determine its operating mode . Table 2.4.1 describes the default settings of the GPHY chip after power-up.

Configure Pin	illustrate	Configuration value
PHYAD[2:0]	PHY address in MDIO/MDC mode	PHY Address is 011
CLK125_EN	Enable 125MHz clock output selection	Enable
LED_MODE	LED light mode configuration	Single LED light mode
MODE0~MODE3	Link adaptive and full-duplex configuration	10/100/1000 auto-negotiation, compatible with full-duplex and half-duplex.

Table 2.4.1 - GPHY Chip Default Settings

When the network is connected to Gigabit Ethernet, the PHY chip KSZ9031RNX communicates via the RGMII bus for data transmission. The transmission clock is 125MHz, and the data is sampled on the rising and falling edges of the clock.

When the network is connected to a 100Mbps Ethernet network, the KSZ9031RNX PHY chip transmits data via the RMII bus with a transmission clock of 25MHz. Data is sampled on both the rising and falling edges of the clock. A schematic diagram of the Ethernet PHY chip connection is shown in Figure 2.4.1 .

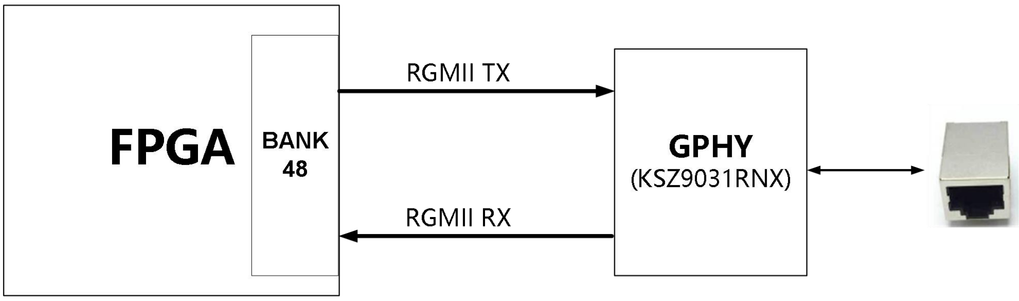

Figure 2.4.1 - Schematic diagram of Ethernet chip connection

The Gigabit Ethernet FPGA pin assignments are as follows:

Signal name	pin name	pin number	Remark
PHY_GTXC	B48_L21_N	W34	Ethernet 1 transmit clock
PHY_TXD0	B48_L18_N	AD33	Ethernet 1 transmits data bit 0
PHY_TXD1	B48_L18_P	AC33	Ethernet 1 transmits data bit 1
PHY_TXD2	B48_L23_N	V34	Ethernet 1 transmits data bit 2
PHY_TXD3	B48_L23_P	U34	Ethernet 1 transmits data bit 3
PHY_TXEN	B48_L21_P	V33	Ethernet 1 sends enable signal
PHY_RXC	B48_L12_P	AC31	Ethernet 1 receive clock
PHY_RXD3	B48_L17_N	AB34	Ethernet 1 Receive Data Bit0
PHY_RXD2	B48_L17_P	AA34	Ethernet 1 Receive Data Bit 1
PHY_RXD1	B48_L15_N	AD34	Ethernet 1 receives data Bit 2
PHY_RXD0	B48_L15_P	AC34	Ethernet 1 receives data Bit 3
PHY_RXDV	B48_L12_N	AC32	Ethernet 1 receive data valid signal
PHY_MDC	B48_T2U	AA33	Ethernet 1MDIO Management Clock
PHY_MDIO	B48_T1U	AE31	Ethernet 1MDIO Management Data
PHY_RESET	B48_T3U	V32	Ethernet chip reset

Table 2.4.1 - Gigabit Ethernet FPGA Pin Assignments

2.5 USB to Serial Port

The AXKU062 development board is equipped with a UART-to-USB interface for serial communication and debugging. The conversion chip is the Silicon Labs CP2102GM USB- UART chip . A level converter chip connects the CP2102 serial chip and the FPGA to accommodate different FPGA bank voltages. The USB interface is a MINI USB interface, which can be connected to the PC 's USB port using a USB cable for serial data communication. A schematic diagram of the USB UART circuit design is shown below :

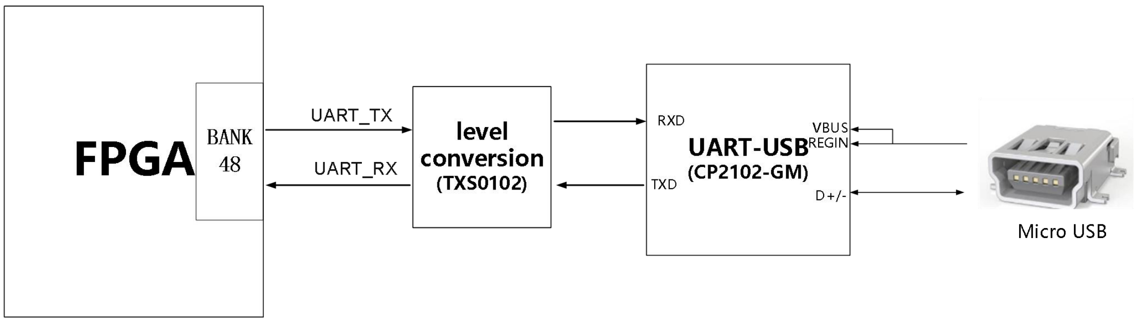

Figure 2.5.1 - Schematic diagram of USB to serial port

FPGA pin assignment for USB to serial port conversion:

Signal name	FPGA pin names	pin number	Remark
UART_RXD	B64_T1U	AJ11	UART data input
UART_TXD	B64_T3U	AM9	UART data output

Table 2.5.1 - USB to Serial Port FPGA Pin Assignment

2.6 FMC Expansion Port

The AXKU062 development board has two FMC LPC expansion ports and one FMC HPC expansion port, which can be connected to various FMC modules from Xilinx or Alinx (HDMI input/output modules, binocular camera modules, high-speed AD modules, etc.).

The LPC FMC1 expansion port has 36 pairs of differential signals, which are connected to the IO pins of BANK47 and BANK48 on the FPGA chip. The IO pin levels of BANK47 and BANK48 are 1.8V. One pair of high-speed GTH transceiver signals is connected to the BNAK226.

The LPC FMC2 expansion port has 36 pairs of differential signals, which are connected to the IO of BANK64 and BANK65 of the FPGA chip respectively, with an IO level of 3.3V.

The FMC HPC expansion port contains 58 pairs of differential I/O signals, which are connected to FPGA chips BANK66, BANK67, and BANK68 respectively, with a voltage standard of 1.8V. Eight high-speed GTH transceiver signals are connected to the I/O pins of FPGA chips BANK227 and BANK228. The schematic diagram of the FPGA and FMC LPC connector is shown below :

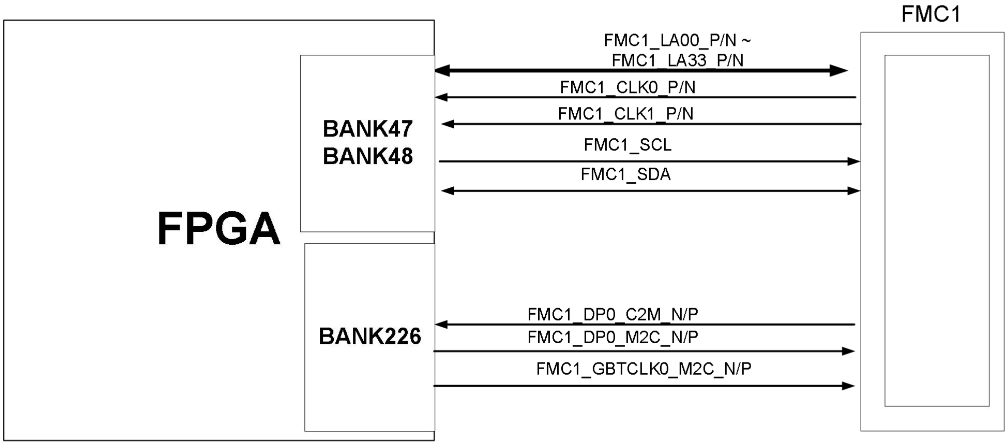

Figure 2.6.1 - Schematic diagram of LPC FMC1 connection

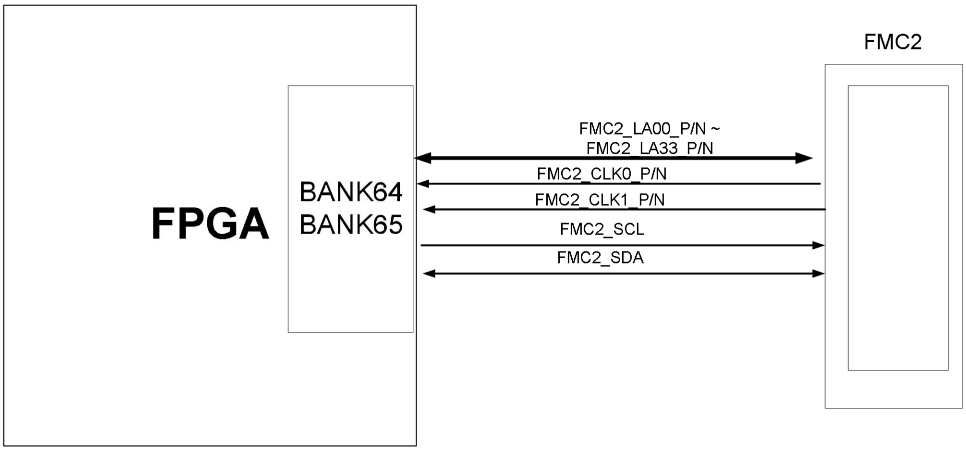

Figure 2.6.2 - Schematic diagram of LPC FMC2 connection

The FPGA and FMC3 HPC connector is shown in Figure 2.6.3 :

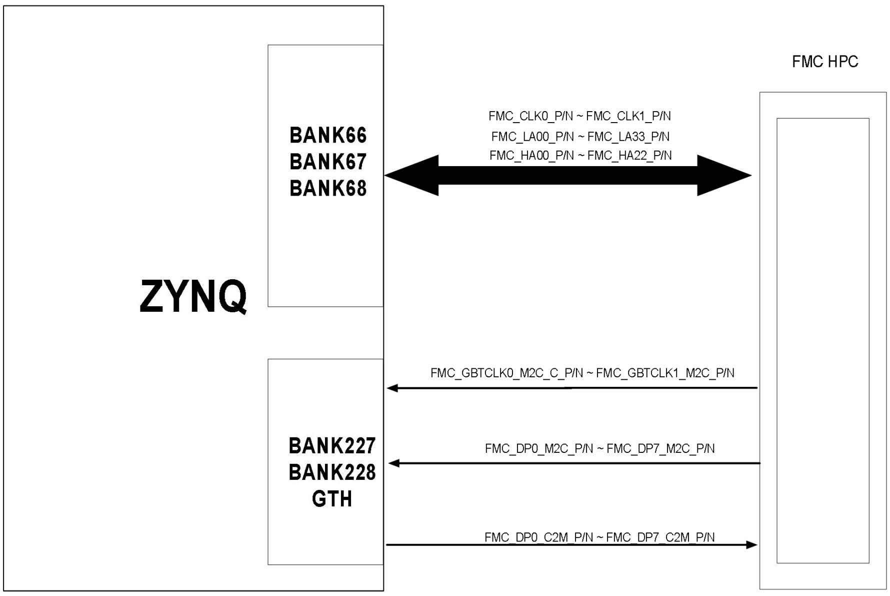

Figure 2.6.3 - HPC FMC3 Connection Diagram

The pin assignments for the first FMC LPC connector are as follows:

Pin No.	Signal name	FPGA pin names	Pin number	Remark
H5	FMC1_LPC_CLK0_N	B47_L11_N	AA23	FMC Reference Clock 1 Negative
H4	FMC1_LPC_CLK0_P	B47_L11_P	Y23	FMC Reference Clock 1 Positive
G3	FMC1_LPC_CLK1_N	B48_L14_N	AB31	FMC Reference Clock 2 Negative
G2	FMC1_LPC_CLK1_P	B48_L14_P	AB30	FMC Reference Clock 2 Positive
G7	FMC1_LPC_LA00_CC_N	B47_L13_N	W24	FMC Reference Data (Clock) Lane 0 Negative
G6	FMC1_LPC_LA00_CC_P	B47_L13_P	W23	FMC Reference Data (Clock) Lane 0 Positive
D9	FMC1_LPC_LA01_CC_N	B47_L12_N	AA25	FMC Reference Data (Clock) Lane 1 Negative
D8	FMC1_LPC_LA01_CC_P	B47_L12_P	AA24	FMC Reference Data (Clock) Lane 1 Positive
H8	FMC1_LPC_LA02_N	B47_L18_N	W21	FMC Reference Data Lane 2 Negative
H7	FMC1_LPC_LA02_P	B47_L18_P	V21	FMC Reference Data Lane 2 Positive
G10	FMC1_LPC_LA03_N	B47_L16_N	V23	FMC Reference Data Lane 3 Negative
G9	FMC1_LPC_LA03_P	B47_L16_P	V22	FMC Reference Data Lane 3 Positive
H11	FMC1_LPC_LA04_N	B47_L6_N	AB26	FMC Reference Data Lane 4 Negative
H10	FMC1_LPC_LA04_P	B47_L6_P	AB25	FMC Reference Data Lane 4 Positive
D12	FMC1_LPC_LA05_N	B47_L23_N	W29	FMC Reference Data Lane 5 Negative
D11	FMC1_LPC_LA05_P	B47_L23_P	V29	FMC Reference Data Lane 5 Positive
C11	FMC1_LPC_LA06_N	B47_L1_N	Y27	FMC Reference Data Lane 6 Negative
C10	FMC1_LPC_LA06_P	B47_L1_P	Y26	FMC Reference Data Lane 6 Positive
H14	FMC1_LPC_LA07_N	B47_L15_N	U22	FMC Reference Data Lane 7 Negative
H13	FMC1_LPC_LA07_P	B47_L15_P	U21	FMC Reference Data Lane 7 Positive
G13	FMC1_LPC_LA08_N	B47_L24_N	W26	FMC Reference Data Lane 8 Negative
G12	FMC1_LPC_LA08_P	B47_L24_P	V26	FMC Reference Data Lane 8 Positive

D15	FMC1_LPC_LA09_N	B47_L17_N	T23	FMC Reference Data Lane 9 Negative
D14	FMC1_LPC_LA09_P	B47_L17_P	T22	FMC Reference Data Lane 9 Positive
C15	FMC1_LPC_LA10_N	B47_L20_N	U25	FMC Reference Data Lane 10 Negative
C14	FMC1_LPC_LA10_P	B47_L20_P	U24	FMC Reference Data Lane 10 Positive
H17	FMC1_LPC_LA11_N	B47_L3_N	AC24	FMC Reference Data Lane 11 Negative
H16	FMC1_LPC_LA11_P	B47_L3_P	AB24	FMC Reference Data Lane 11 Positive
G16	FMC1_LPC_LA12_N	B47_L22_N	U27	FMC Reference Data Lane 12 Negative
G15	FMC1_LPC_LA12_P	B47_L22_P	U26	FMC Reference Data Lane 12 Positive
D18	FMC1_LPC_LA13_N	B47_L21_N	Y28	FMC Reference Data Lane 13 Negative
D17	FMC1_LPC_LA13_P	B47_L21_P	W28	FMC Reference Data Lane 13 Positive
C19	FMC1_LPC_LA14_N	B47_L19_N	V28	FMC Reference Data Lane 14 Negative
C18	FMC1_LPC_LA14_P	B47_L19_P	V27	FMC Reference Data Lane 14 Positive
H20	FMC1_LPC_LA15_N	B47_L14_N	Y25	FMC Reference Data Lane 15 Negative
H19	FMC1_LPC_LA15_P	B47_L14_P	W25	FMC Reference Data Lane 15 Positive
G19	FMC1_LPC_LA16_N	B47_L7_N	AB22	FMC Reference Data Lane 16 Negative
G18	FMC1_LPC_LA16_P	B47_L7_P	AA22	FMC Reference Data Lane 16 Positive
D21	FMC1_LPC_LA17_CC_N	B48_L13_N	AB32	FMC Reference Data (Clock) Lane 17 Negative
D20	FMC1_LPC_LA17_CC_P	B48_L13_P	AA32	FMC Reference Data (Clock) Lane 17 Positive
C23	FMC1_LPC_LA18_CC_N	B48_L11_N	AD31	FMC Reference Data (Clock) Lane 18 Negative
C22	FMC1_LPC_LA18_CC_P	B48_L11_P	AD30	FMC Reference Data (Clock) Lane 18 Positive
H23	FMC1_LPC_LA19_N	B48_L16_N	AB29	FMC Reference Data Lane 19 Negative
H22	FMC1_LPC_LA19_P	B48_L16_P	AA29	FMC Reference Data Lane 19 Positive
G22	FMC1_LPC_LA20_N	B48_L24_N	W31	FMC Reference Data Lane 20 Negative
G21	FMC1_LPC_LA20_P	B48_L24_P	V31	FMC Reference Data Lane 20 Positive

H26	FMC1_LPC_LA21_N	B48_L6_N	AG30	FMC Reference Data Lane 21 Negative
H25	FMC1_LPC_LA21_P	B48_L6_P	AF30	FMC Reference Data Lane 21 Positive
G25	FMC1_LPC_LA22_N	B48_L5_N	AE30	FMC Reference Data Lane 22 Negative
G24	FMC1_LPC_LA22_P	B48_L5_P	AD29	FMC Reference Data Lane 22 Positive
D24	FMC1_LPC_LA23_N	B48_L8_N	AG34	FMC Reference Data Lane 23 Negative
D23	FMC1_LPC_LA23_P	B48_L8_P	AF33	FMC Reference Data Lane 23 Positive
H29	FMC1_LPC_LA24_N	B48_L4_N	AG29	FMC Reference Data Lane 24 Negative
H28	FMC1_LPC_LA24_P	B48_L4_P	AF29	FMC Reference Data Lane 24 Positive
G28	FMC1_LPC_LA25_N	B48_L9_N	AF32	FMC Reference Data Lane 25 Negative
G27	FMC1_LPC_LA25_P	B48_L9_P	AE32	FMC Reference Data Lane 25 Positive
D27	FMC1_LPC_LA26_N	B48_L7_N	AG32	FMC Reference Data Lane 26 Negative
D26	FMC1_LPC_LA26_P	B48_L7_P	AG31	FMC Reference Data Lane 26 Positive
C27	FMC1_LPC_LA27_N	B48_L10_N	AF34	FMC Reference Data Lane 27 Negative
C26	FMC1_LPC_LA27_P	B48_L10_P	AE33	FMC Reference Data Lane 27 Positive
H32	FMC1_LPC_LA28_N	B48_L1_N	AF27	FMC Reference Data Lane 28 Negative
H31	FMC1_LPC_LA28_P	B48_L1_P	AE27	FMC Reference Data Lane 28 Positive
G31	FMC1_LPC_LA29_N	B48_L2_N	AF28	FMC Reference Data Lane 29 Negative
G30	FMC1_LPC_LA29_P	B48_L2_P	AE28	FMC Reference Data Lane 29 Positive
H35	FMC1_LPC_LA30_N	B48_L3_N	AD28	FMC Reference Data Lane 30 Negative
H34	FMC1_LPC_LA30_P	B48_L3_P	AC28	FMC Reference Data Lane 30 Positive
G34	FMC1_LPC_LA31_N	B48_L19_N	Y33	FMC Reference Data Lane 31 Negative
G33	FMC1_LPC_LA31_P	B48_L19_P	W33	FMC Reference Data Lane 31 Positive
H38	FMC1_LPC_LA32_N	B48_L22_N	Y32	FMC Reference Data Lane 32 Negative
H37	FMC1_LPC_LA32_P	B48_L22_P	Y31	FMC Reference Data Lane 32 Positive

G37	FMC1_LPC_LA33_N	B48_L20_N	Y30	FMC Reference Data Lane 33 Negative
G36	FMC1_LPC_LA33_P	B48_L20_P	W30	FMC Reference Data Lane 33 Positive
C30	FMC1_LPC_SCL	B47_L10_N	AC21	FMC I2C Bus Clock
C31	FMC1_LPC_SDA	B47_L10_P	AB21	FMC I2C Bus Data
C3	FMC1_DP0_C2M_N	226_TX3_N	R3	Transceiver Data Output Negative
C2	FMC1_DP0_C2M_P	226_TX3_P	R4	Transceiver Data Output Positive
C7	FMC1_DP0_M2C_N	226_RX3_N	P1	Transceiver Data Input Negative
C6	FMC1_DP0_M2C_P	226_RX3_P	P2	Transceiver Data Input Positive
D5	FMC1_GBTCLK0_M2C_N	226_CLK0_N	V5	Transceiver Reference Clock Negative
D4	FMC1_GBTCLK0_M2C_P	226_CLK0_P	V6	Transceiver Reference Clock Positive

Table 2.6.1 - Pin Assignment for the First FMC LPC Connector

The pin assignments for the second FMC LPC connector are as follows:

pin No.	Signal name	pin name	pin number	Remark
H5	FMC2_LPC_CLK0_N	B65_L13_N	N26	FMC Reference Clock 1 Negative
H4	FMC2_LPC_CLK0_P	B65_L13_P	P26	FMC Reference Clock 1 Positive
G3	FMC2_LPC_CLK1_N	B64_L11_N	AH12	FMC Reference Clock 2 Negative
G2	FMC2_LPC_CLK1_P	B64_L11_P	AG12	FMC Reference Clock 2 Positive
G7	FMC2_LPC_LA00_CC_N	B65_L14_N	P25	FMC Reference Data (Clock) Lane 0 Negative
G6	FMC2_LPC_LA00_CC_P	B65_L14_P	P24	FMC Reference Data (Clock) Lane 0 Positive
D9	FMC2_LPC_LA01_CC_N	B65_L11_N	M26	FMC Reference Data (Clock) Lane 1 Negative
D8	FMC2_LPC_LA01_CC_P	B65_L11_P	M25	FMC Reference Data (Clock) Lane 1 Positive
H8	FMC2_LPC_LA02_N	B65_L17_N	R26	FMC Reference Data Lane 2 Negative
H7	FMC2_LPC_LA02_P	B65_L17_P	R25	FMC Reference Data Lane 2 Positive
G10	FMC2_LPC_LA03_N	B65_L7_N	L27	FMC Reference Data Lane 3 Negative

G9	FMC2_LPC_LA03_P	B65_L7_P	M27	FMC Reference Data Lane 3 Positive
H11	FMC2_LPC_LA04_N	B65_L15_N	R27	FMC Reference Data Lane 4 Negative
H10	FMC2_LPC_LA04_P	B65_L15_P	T27	FMC Reference Data Lane 4 Positive
D12	FMC2_LPC_LA05_N	B65_L4_N	J25	FMC Reference Data Lane 5 Negative
D11	FMC2_LPC_LA05_P	B65_L4_P	J24	FMC Reference Data Lane 5 Positive
C11	FMC2_LPC_LA06_N	B65_L3_N	K27	FMC Reference Data Lane 6 Negative
C10	FMC2_LPC_LA06_P	B65_L3_P	K26	FMC Reference Data Lane 6 Positive
H14	FMC2_LPC_LA07_N	B65_L5_N	H26	FMC Reference Data Lane 7 Negative
H13	FMC2_LPC_LA07_P	B65_L5_P	J26	FMC Reference Data Lane 7 Positive
G13	FMC2_LPC_LA08_N	B65_L18_N	P23	FMC Reference Data Lane 8 Negative
G12	FMC2_LPC_LA08_P	B65_L18_P	R23	FMC Reference Data Lane 8 Positive
D15	FMC2_LPC_LA09_N	B65_L1_N	G27	FMC Reference Data Lane 9 Negative
D14	FMC2_LPC_LA09_P	B65_L1_P	H27	FMC Reference Data Lane 9 Positive
C15	FMC2_LPC_LA10_N	B65_L20_N	P21	FMC Reference Data Lane 10 Negative
C14	FMC2_LPC_LA10_P	B65_L20_P	P20	FMC Reference Data Lane 10 Positive
H17	FMC2_LPC_LA11_N	B65_L9_N	K25	FMC Reference Data Lane 11 Negative
H16	FMC2_LPC_LA11_P	B65_L9_P	L25	FMC Reference Data Lane 11 Positive
G16	FMC2_LPC_LA12_N	B65_L12_N	M24	FMC Reference Data Lane 12 Negative
G15	FMC2_LPC_LA12_P	B65_L12_P	N24	FMC Reference Data Lane 12 Positive
D18	FMC2_LPC_LA13_N	B65_L19_N	M22	FMC Reference Data Lane 13 Negative
D17	FMC2_LPC_LA13_P	B65_L19_P	N22	FMC Reference Data Lane 13 Positive
C19	FMC2_LPC_LA14_N	B65_L23_N	M21	FMC Reference Data Lane 14 Negative
C18	FMC2_LPC_LA14_P	B65_L23_P	N21	FMC Reference Data Lane 14 Positive
H20	FMC2_LPC_LA15_N	B65_L10_N	K23	FMC Reference Data Lane 15 Negative

H19	FMC2_LPC_LA15_P	B65_L10_P	L22	FMC Reference Data Lane 15 Positive
G19	FMC2_LPC_LA16_N	B65_L6_N	H24	FMC Reference Data Lane 16 Negative
G18	FMC2_LPC_LA16_P	B65_L6_P	J23	FMC Reference Data Lane 16 Positive
D21	FMC2_LPC_LA17_CC_N	B64_L13_N	AG10	FMC Reference Data (Clock) Lane 17 Negative
D20	FMC2_LPC_LA17_CC_P	B64_L13_P	AF10	FMC Reference Data (Clock) Lane 17 Positive
C23	FMC2_LPC_LA18_CC_N	B64_L12_N	AH11	FMC Reference Data (Clock) Lane 18 Negative
C22	FMC2_LPC_LA18_CC_P	B64_L12_P	AG11	FMC Reference Data (Clock) Lane 18 Positive
H23	FMC2_LPC_LA19_N	B64_L17_N	AD8	FMC Reference Data Lane 19 Negative
H22	FMC2_LPC_LA19_P	B64_L17_P	AD9	FMC Reference Data Lane 19 Positive
G22	FMC2_LPC_LA20_N	B64_L23_N	AJ8	FMC Reference Data Lane 20 Negative
G21	FMC2_LPC_LA20_P	B64_L23_P	AJ9	FMC Reference Data Lane 20 Positive
H26	FMC2_LPC_LA21_N	B64_L14_N	AG9	FMC Reference Data Lane 21 Negative
H25	FMC2_LPC_LA21_P	B64_L14_P	AF9	FMC Reference Data Lane 21 Positive
G25	FMC2_LPC_LA22_N	B64_L15_N	AF8	FMC Reference Data Lane 22 Negative
G24	FMC2_LPC_LA22_P	B64_L15_P	AE8	FMC Reference Data Lane 22 Positive
D24	FMC2_LPC_LA23_N	B64_L16_N	AE10	FMC Reference Data Lane 23 Negative
D23	FMC2_LPC_LA23_P	B64_L16_P	AD10	FMC Reference Data Lane 23 Positive
H29	FMC2_LPC_LA24_N	B64_L1_N	AP10	FMC Reference Data Lane 24 Negative
H28	FMC2_LPC_LA24_P	B64_L1_P	AP11	FMC Reference Data Lane 24 Positive
G28	FMC2_LPC_LA25_N	B64_L4_N	AN12	FMC Reference Data Lane 25 Negative
G27	FMC2_LPC_LA25_P	B64_L4_P	AM12	FMC Reference Data Lane 25 Positive
D27	FMC2_LPC_LA26_N	B64_L21_N	AL9	FMC Reference Data Lane 26 Negative
D26	FMC2_LPC_LA26_P	B64_L21_P	AK10	FMC Reference Data Lane 26 Positive
C27	FMC2_LPC_LA27_N	B64_L24_N	AL8	FMC Reference Data Lane 27 Negative

C26	FMC2_LPC_LA27_P	B64_L24_P	AK8	FMC Reference Data Lane 27 Positive
H32	FMC2_LPC_LA28_N	B64_L18_N	AH8	FMC Reference Data Lane 28 Negative
H31	FMC2_LPC_LA28_P	B64_L18_P	AH9	FMC Reference Data Lane 28 Positive
G31	FMC2_LPC_LA29_N	B64_L6_N	AL13	FMC Reference Data Lane 29 Negative
G30	FMC2_LPC_LA29_P	B64_L6_P	AK13	FMC Reference Data Lane 29 Positive
H35	FMC2_LPC_LA30_N	B64_L8_N	AJ13	FMC Reference Data Lane 30 Negative
H34	FMC2_LPC_LA30_P	B64_L8_P	AH13	FMC Reference Data Lane 30 Positive
G34	FMC2_LPC_LA31_N	B64_L10_N	AE11	FMC Reference Data Lane 31 Negative
G33	FMC2_LPC_LA31_P	B64_L10_P	AD11	FMC Reference Data Lane 31 Positive
H38	FMC2_LPC_LA32_N	B64_L7_N	AF13	FMC Reference Data Lane 32 Negative
H37	FMC2_LPC_LA32_P	B64_L7_P	AE13	FMC Reference Data Lane 32 Positive
G37	FMC2_LPC_LA33_N	B64_L9_N	AF12	FMC Reference Data Lane 33 Negative
G36	FMC2_LPC_LA33_P	B64_L9_P	AE12	FMC Reference Data Lane 33 Positive
C30	FMC2_LPC_SCL	B65_L24_N	K21	FMC I2C Bus Clock
C31	FMC2_LPC_SDA	B65_L24_P	K20	FMC I2C Bus Data

Table 2.6.2 - Pin Assignment for the Second FMC LPC Connector

The pin assignments for the third FMC HPC connector are as follows:

pin No.	signal name	FPGA pin names	pin number	Remark
H5	FMC_HPC_CLK0_M2C_N	B67_L11_N	D25	FMC Input Reference Clock 0 Negative
H4	FMC_HPC_CLK0_M2C_P	B67_L11_P	E25	FMC Input Reference Clock 0 Positive
G3	FMC_HPC_CLK1_M2C_N	B66_L13_N	G11	FMC Input Reference Clock 1 Negative
G2	FMC_HPC_CLK1_M2C_P	B66_L13_P	H11	FMC Input Reference Clock 1 Positive
G7	FMC_HPC_LA00_CC_N	B67_L14_N	E23	FMC LA Data (Clock) Lane 0 Negative
G6	FMC_HPC_LA00_CC_P	B67_L14_P	E22	FMC LA Data (Clock) Lane 0 Positive

D9	FMC_HPC_LA01_CC_N	B67_L13_N	C23	FMC LA Data (Clock) Lane 1 Negative
D8	FMC_HPC_LA01_CC_P	B67_L13_P	D23	FMC LA Data (Clock) Lane 1 Positive
H8	FMC_HPC_LA02_N	B67_L8_N	A25	FMC LA Data Lane 2 Negative
H7	FMC_HPC_LA02_P	B67_L8_P	B25	FMC LA Data Lane 2 Positive
G10	FMC_HPC_LA03_N	B67_L6_N	A28	FMC LA Data Lane 3 Negative
G9	FMC_HPC_LA03_P	B67_L6_P	A27	FMC LA Data Lane 3 Positive
H11	FMC_HPC_LA04_N	B67_L2_N	B27	FMC LA Data Lane 4 Negative
H10	FMC_HPC_LA04_P	B67_L2_P	C27	FMC LA Data Lane 4 Positive
D12	FMC_HPC_LA05_N	B67_L12_N	C24	FMC LA Data Lane 5 Negative
D11	FMC_HPC_LA05_P	B67_L12_P	D24	FMC LA Data Lane 5 Positive
C11	FMC_HPC_LA06_N	B67_L4_N	A29	FMC LA Data Lane 6 Negative
C10	FMC_HPC_LA06_P	B67_L4_P	B29	FMC LA Data Lane 6 Positive
H14	FMC_HPC_LA07_N	B67_L5_N	C28	FMC LA Data Lane 7 Negative
H13	FMC_HPC_LA07_P	B67_L5_P	D28	FMC LA Data Lane 7 Positive
G13	FMC_HPC_LA08_N	B67_L1_N	E27	FMC LA Data Lane 8 Negative
G12	FMC_HPC_LA08_P	B67_L1_P	F27	FMC LA Data Lane 8 Positive
D15	FMC_HPC_LA09_N	B67_L9_N	B26	FMC LA Data Lane 9 Negative
D14	FMC_HPC_LA09_P	B67_L9_P	C26	FMC LA Data Lane 9 Positive
C15	FMC_HPC_LA10_N	B67_L10_N	A24	FMC LA Data Lane 10 Negative
C14	FMC_HPC_LA10_P	B67_L10_P	B24	FMC LA Data Lane 10 Positive
H17	FMC_HPC_LA11_N	B67_L7_N	D26	FMC LA Data Lane 11 Negative
H16	FMC_HPC_LA11_P	B67_L7_P	E26	FMC LA Data Lane 11 Positive
G16	FMC_HPC_LA12_N	B67_L3_N	D29	FMC LA Data Lane 12 Negative
G15	FMC_HPC_LA12_P	B67_L3_P	E28	FMC LA Data Lane 12 Positive

D18	FMC_HPC_LA13_N	B67_L15_N	B22	FMC LA Data Lane 13 Negative
D17	FMC_HPC_LA13_P	B67_L15_P	B21	FMC LA Data Lane 13 Positive
C19	FMC_HPC_LA14_N	B67_L18_N	D21	FMC LA Data Lane 14 Negative
C18	FMC_HPC_LA14_P	B67_L18_P	D20	FMC LA Data Lane 14 Positive
H20	FMC_HPC_LA15_N	B67_L17_N	A20	FMC LA Data Lane 15 Negative
H19	FMC_HPC_LA15_P	B67_L17_P	B20	FMC LA Data Lane 15 Positive
G19	FMC_HPC_LA16_N	B67_L16_N	C22	FMC LA Data Lane 16 Negative
G18	FMC_HPC_LA16_P	B67_L16_P	C21	FMC LA Data Lane 16 Positive
D21	FMC_HPC_LA17_CC_N	B66_L11_N	F9	FMC LA Data (Clock) Lane 17 Negative
D20	FMC_HPC_LA17_CC_P	B66_L11_P	G9	FMC LA Data (Clock) Lane 17 Positive
C23	FMC_HPC_LA18_CC_N	B66_L12_N	F10	FMC LA Data (Clock) Lane 18 Negative
C22	FMC_HPC_LA18_CC_P	B66_L12_P	G10	FMC LA Data (Clock) Lane 18 Positive
H23	FMC_HPC_LA19_N	B66_L21_N	B11	FMC LA Data Lane 19 Negative
H22	FMC_HPC_LA19_P	B66_L21_P	C11	FMC LA Data Lane 19 Positive
G22	FMC_HPC_LA20_N	B66_L23_N	A12	FMC LA Data Lane 20 Negative
G21	FMC_HPC_LA20_P	B66_L23_P	A13	FMC LA Data Lane 20 Positive
H26	FMC_HPC_LA21_N	B66_L15_N	J11	FMC LA Data Lane 21 Negative
H25	FMC_HPC_LA21_P	B66_L15_P	K11	FMC LA Data Lane 21 Positive
G25	FMC_HPC_LA22_N	B66_L19_N	D11	FMC LA Data Lane 22 Negative
G24	FMC_HPC_LA22_P	B66_L19_P	E11	FMC LA Data Lane 22 Positive
D24	FMC_HPC_LA23_N	B66_L18_N	H13	FMC LA Data Lane 23 Negative
D23	FMC_HPC_LA23_P	B66_L18_P	J13	FMC LA Data Lane 23 Positive
H29	FMC_HPC_LA24_N	B66_L8_N	H9	FMC LA Data Lane 24 Negative
H28	FMC_HPC_LA24_P	B66_L8_P	J9	FMC LA Data Lane 24 Positive

G28	FMC_HPC_LA25_N	B66_L10_N	J10	FMC LA Data Lane 25 Negative
G27	FMC_HPC_LA25_P	B66_L10_P	K10	FMC LA Data Lane 25 Positive
D27	FMC_HPC_LA26_N	B66_L6_N	D10	FMC LA Data Lane 26 Negative
D26	FMC_HPC_LA26_P	B66_L6_P	E10	FMC LA Data Lane 26 Positive
C27	FMC_HPC_LA27_N	B66_L5_N	C9	FMC LA Data Lane 27 Negative
C26	FMC_HPC_LA27_P	B66_L5_P	D9	FMC LA Data Lane 27 Positive
H32	FMC_HPC_LA28_N	B66_L2_N	A9	FMC LA Data Lane 28 Negative
H31	FMC_HPC_LA28_P	B66_L2_P	B9	FMC LA Data Lane 28 Positive
G31	FMC_HPC_LA29_N	B66_L4_N	A10	FMC LA Data Lane 29 Negative
G30	FMC_HPC_LA29_P	B66_L4_P	B10	FMC LA Data Lane 29 Positive
H35	FMC_HPC_LA30_N	B66_L9_N	H8	FMC LA Data Lane 30 Negative
H34	FMC_HPC_LA30_P	B66_L9_P	J8	FMC LA Data Lane 30 Positive
G34	FMC_HPC_LA31_N	B66_L1_N	E8	FMC LA Data Lane 31 Negative
G33	FMC_HPC_LA31_P	B66_L1_P	F8	FMC LA Data Lane 31 Positive
H38	FMC_HPC_LA32_N	B66_L3_N	C8	FMC LA Data Lane 32 Negative
H37	FMC_HPC_LA32_P	B66_L3_P	D8	FMC LA Data Lane 32 Positive
G37	FMC_HPC_LA33_N	B66_L7_N	K8	FMC LA Data Lane 33 Negative
G36	FMC_HPC_LA33_P	B66_L7_P	L8	FMC LA Data Lane 33 Positive
F5	FMC_HPC_HA00_CC_N	B68_L14_N	F17	FMC HA Data (Clock) Lane 0 Negative
F4	FMC_HPC_HA00_CC_P	B68_L14_P	F18	FMC HA Data (Clock) Lane 0 Positive
E3	FMC_HPC_HA01_CC_N	B68_L12_N	E17	FMC HA Data (Clock) Lane 1 Negative
E2	FMC_HPC_HA01_CC_P	B68_L12_P	E18	FMC HA Data (Clock) Lane 1 Positive
K8	FMC_HPC_HA02_N	B68_L17_N	H16	FMC HA Data Lane 2 Negative
K7	FMC_HPC_HA02_P	B68_L17_P	H17	FMC HA Data Lane 2 Positive

J7	FMC_HPC_HA03_N	B68_L24_N	L18	FMC HA Data Lane 3 Negative
J6	FMC_HPC_HA03_P	B68_L24_P	L19	FMC HA Data Lane 3 Positive
F8	FMC_HPC_HA04_N	B68_L6_N	C17	FMC HA Data Lane 4 Negative
F7	FMC_HPC_HA04_P	B68_L6_P	C18	FMC HA Data Lane 4 Positive
E7	FMC_HPC_HA05_N	B68_L2_N	A18	FMC HA Data Lane 5 Negative
E6	FMC_HPC_HA05_P	B68_L2_P	A19	FMC HA Data Lane 5 Positive
K11	FMC_HPC_HA06_N	B68_L22_N	J18	FMC HA Data Lane 6 Negative
K10	FMC_HPC_HA06_P	B68_L22_P	J19	FMC HA Data Lane 6 Positive
J10	FMC_HPC_HA07_N	B68_L4_N	B19	FMC HA Data Lane 7 Negative
J9	FMC_HPC_HA07_P	B68_L4_P	C19	FMC HA Data Lane 7 Positive
F11	FMC_HPC_HA08_N	B68_L18_N	H18	FMC HA Data Lane 8 Negative
F10	FMC_HPC_HA08_P	B68_L18_P	H19	FMC HA Data Lane 8 Positive
E10	FMC_HPC_HA09_N	B68_L7_N	C14	FMC HA Data Lane 9 Negative
E9	FMC_HPC_HA09_P	B68_L7_P	D14	FMC HA Data Lane 9 Positive
K14	FMC_HPC_HA10_N	B68_L1_N	A14	FMC HA Data Lane 10 Negative
K13	FMC_HPC_HA10_P	B68_L1_P	B14	FMC HA Data Lane 10 Positive
J13	FMC_HPC_HA11_N	B68_L5_N	B16	FMC HA Data Lane 11 Negative
J12	FMC_HPC_HA11_P	B68_L5_P	B17	FMC HA Data Lane 11 Positive
F14	FMC_HPC_HA12_N	B68_L16_N	F19	FMC HA Data Lane 12 Negative
F13	FMC_HPC_HA12_P	B68_L16_P	G19	FMC HA Data Lane 12 Positive
E13	FMC_HPC_HA13_N	B68_L3_N	A15	FMC HA Data Lane 13 Negative
E12	FMC_HPC_HA13_P	B68_L3_P	B15	FMC HA Data Lane 13 Positive
J16	FMC_HPC_HA14_N	B68_L23_N	J16	FMC HA Data Lane 14 Negative
J15	FMC_HPC_HA14_P	B68_L23_P	K16	FMC HA Data Lane 14 Positive

J17	FMC_HPC_HA15_N	B68_L20_N	K17	FMC HA Data Lane 15 Negative
F16	FMC_HPC_HA15_P	B68_L20_P	K18	FMC HA Data Lane 15 Positive
F16	FMC_HPC_HA16_N	B68_L10_N	D18	FMC HA Data Lane 16 Negative
E15	FMC_HPC_HA16_P	B68_L10_P	D19	FMC HA Data Lane 16 Positive
K17	FMC_HPC_HA17_CC_N	B68_L13_N	G16	FMC HA Data (Clock) Lane 17 Negative
K16	FMC_HPC_HA17_CC_P	B68_L13_P	G17	FMC HA Data (Clock) Lane 17 Positive
J19	FMC_HPC_HA18_N	B68_L21_N	K15	FMC HA Data Lane 18 Negative
J18	FMC_HPC_HA18_P	B68_L21_P	L15	FMC HA Data Lane 18 Positive
F20	FMC_HPC_HA19_N	B68_L15_N	G14	FMC HA Data Lane 19 Negative
F19	FMC_HPC_HA19_P	B68_L15_P	G15	FMC HA Data Lane 19 Positive
E19	FMC_HPC_HA20_N	B68_L11_N	D16	FMC HA Data Lane 20 Negative
E18	FMC_HPC_HA20_P	B68_L11_P	E16	FMC HA Data Lane 20 Positive
K20	FMC_HPC_HA21_N	B68_L19_N	J14	FMC HA Data Lane 21 Negative
K19	FMC_HPC_HA21_P	B68_L19_P	J15	FMC HA Data Lane 21 Positive
J21	FMC_HPC_HA22_N	B68_L8_N	D15	FMC HA Data Lane 22 Negative
J22	FMC_HPC_HA22_P	B68_L8_P	E15	FMC HA Data Lane 22 Positive
K23	FMC_HPC_HA23_N	B68_L9_N	F14	FMC HA Data Lane 23 Negative
K22	FMC_HPC_HA23_P	B68_L9_P	F15	FMC HA Data Lane 23 Positive
C30	FMC_HPC_SCL	B66_L17_N	K12	FMC I2C Bus Clock
C31	FMC_HPC_SDA	B66_L17_P	L12	FMC I2C Bus Data
D4	FMC_GBTCLK0_M2C_P	227_CLK1_P	M6	Transceiver Reference Clock 0 Input Positive
D5	FMC_GBTCLK0_M2C_N	227_CLK1_N	M5	Transceiver Reference Clock 0 Input Negative
B20	FMC_GBTCLK1_M2C_P	228_CLK1_P	H6	Transceiver Reference Clock 1 Input Positive
B21	FMC_GBTCLK1_M2C_N	228_CLK1_N	H5	Transceiver Reference Clock 1 Input Negative

C6	FMC_DP0_M2C_P	227_RX0_P	M2	Transceiver Data 0 Input Positive
C7	FMC_DP0_M2C_N	227_RX0_N	M1	Transceiver Data 0 Input Negative
A2	FMC_DP1_M2C_P	227_RX1_P	K2	Transceiver Data 1 Input Positive
A3	FMC_DP1_M2C_N	227_RX1_N	K1	Transceiver Data 1 Input Negative
A6	FMC_DP2_M2C_P	227_RX2_P	H2	Transceiver Data 2 Input Positive
A7	FMC_DP2_M2C_N	227_RX2_N	H1	Transceiver Data 2 Input Negative
A10	FMC_DP3_M2C_P	227_RX3_P	F2	Transceiver Data 3 Input Positive
A11	FMC_DP3_M2C_N	227_RX3_N	F1	Transceiver Data 3 Input Negative
A14	FMC_DP4_M2C_P	228_RX1_P	D2	Transceiver Data 4 Input Positive
A15	FMC_DP4_M2C_N	228_RX1_N	D1	Transceiver Data 4 Input Negative
A18	FMC_DP5_M2C_P	228_RX3_P	A4	Transceiver Data 5 Input Positive
A19	FMC_DP5_M2C_N	228_RX3_N	A3	Transceiver Data 5 Input Negative
B12	FMC_DP6_M2C_P	228_RX2_P	B2	Transceiver Data 6 Input Positive
B13	FMC_DP6_M2C_N	228_RX2_N	B1	Transceiver Data 6 Input Negative
B16	FMC_DP7_M2C_P	228_RX0_P	E4	Transceiver Data 7 Input Positive
B17	FMC_DP7_M2C_N	228_RX0_N	E3	Transceiver Data 7 Input Negative
C2	FMC_DP0_C2M_P	227_TX0_P	N4	Transceiver Data 0 Output Positive
C3	FMC_DP0_C2M_N	227_TX0_N	N3	Transceiver Data 0 Output Negative
A22	FMC_DP1_C2M_P	227_TX1_P	L4	Transceiver Data 1 Output Positive
A23	FMC_DP1_C2M_N	227_TX1_N	L3	Transceiver Data 1 Output Negative
A26	FMC_DP2_C2M_P	227_TX2_P	J4	Transceiver Data 2 Output Positive
A27	FMC_DP2_C2M_N	227_TX2_N	J3	Transceiver Data 2 Output Negative
A30	FMC_DP3_C2M_P	227_TX3_P	G4	Transceiver Data 3 Output Positive
A31	FMC_DP3_C2M_N	227_TX3_N	G3	Transceiver Data 3 Output Negative

A34	FMC_DP4_C2M_P	228_TX1_P	D6	Transceiver Data 4 Output Positive
A35	FMC_DP4_C2M_N	228_TX1_N	D5	Transceiver Data 4 Output Negative
A38	FMC_DP5_C2M_P	228_TX3_P	B6	Transceiver Data 5 Output Positive
A39	FMC_DP5_C2M_N	228_TX3_N	B5	Transceiver Data 5 Output Negative
B36	FMC_DP6_C2M_P	228_TX2_P	C4	Transceiver Data 6 Output Positive
B37	FMC_DP6_C2M_N	228_TX2_N	C3	Transceiver Data 6 Output Negative
B32	FMC_DP7_C2M_P	228_TX0_P	F6	Transceiver Data 7 Output Positive
B33	FMC_DP7_C2M_N	228_TX0_N	F5	Transceiver Data 7 Output Negative

Table 2.6.3 - Pin Assignment for the 3rd FMC HPC Connector

2.7 SD Card Slot

The AXKU062 development board includes a Micro SD card interface, providing users with access to SD card storage for storing pictures, music, or other user data files.

The signal is connected to the IO signal of BANK 64 of the FPGA . The schematic diagram of the FPGA and SD card connector is shown in Figure 2.7.1 .

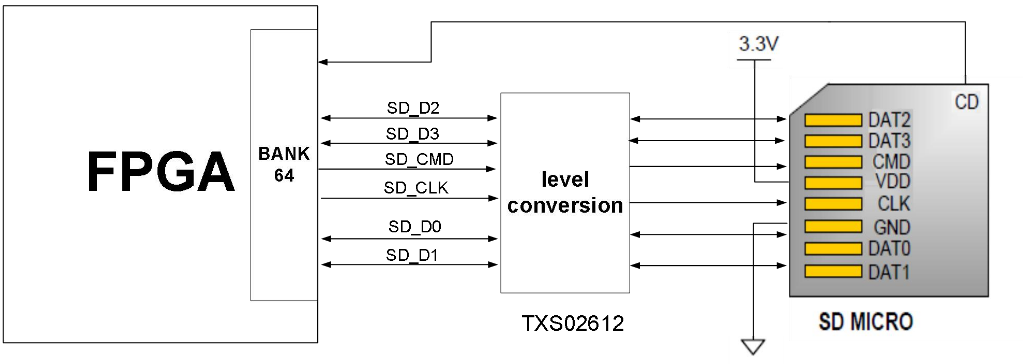

Figure 2.7.1 - SD Card Connection Diagram

SD card slot pin assignment

Signal name	FPGA pin names	FPGA pin numbers	Remark
SD_CLK	B64_L22_P	AN8	SD clock signal
SD_CMD	B64_L19_N	AM10	SD command signals

SD_D0	B64_L5_N	AL12	SD Data0
SD_D1	B64_L19_P	AL10	SD Data1
SD_D2	B64_L2_P	AN13	SD Data2
SD_D3	B64_L2_N	AP13	SD Data3
SD_CD	B64_L22_N	AP8	SD card insertion signal

Table 2.7.1 - SD Card Slot Pin Assignment

2.8 SMA Interface

The AXKU062 development board provides two SMA interfaces , with differential signals connected to the BANK66 general-purpose clock I/O port. This provides an external clock interface or can be viewed as a general-purpose I/O port; the interface voltage is 1.8V. A schematic diagram of the FPGA and SMA interface connection is shown in Figure 2.8.1 .

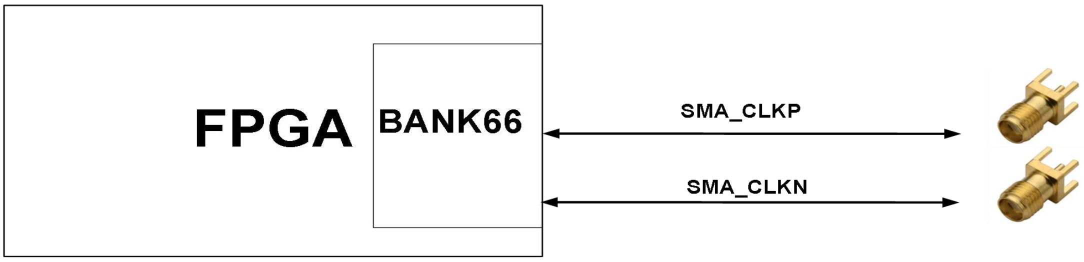

Figure 2.8.1 - Schematic diagram of SMA connector

SMA Interface Pin Assignment

Signal name	FPGA pin names	FPGA pin numbers	Remark
SMA_CLKIN_N	B66_L14_N	G12	Transceiver clock signal N
SMA_CLKIN_P	B66_L14_P	H12	Transceiver clock signal P

Table 2.8.1 - SMA Interface Pin Assignment

2.9 Temperature Sensor and EEPROM

LM75A from ON Semiconductor . The LM75A chip has a temperature accuracy of 0.5 degrees Celsius. The sensor and FPGA share a direct I2C digital interface, allowing the FPGA to read the temperature near the development board. A 4KB EEPROM, model 24LC04, is also connected to the I2C bus.

LM75 sensor and EEPROM chip is shown in Figure 2.9.1.

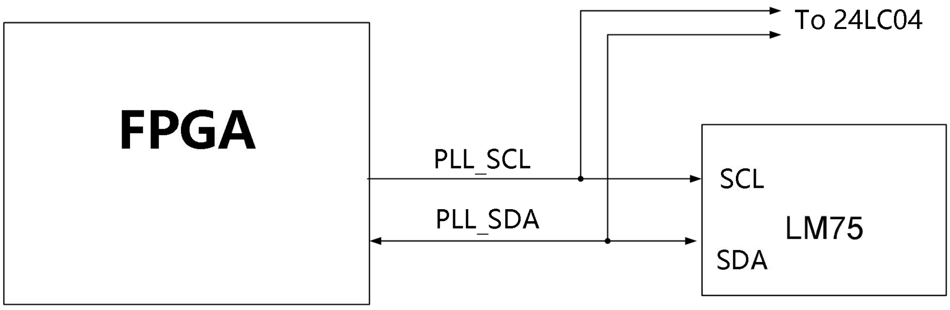

Figure 2.9.1 - Design schematic of LM75 sensor and EEPROM chip

I2C pin assignment:

Pin Name	FPGA pin names	FPGA pins
I2C_SDA	B66_L16_N	K13
I2C_SCL	B66_L16_P	L13

Table 2.9.1 - I2C Pin Assignment

2.10 LED Lights

The AXKU062 baseboard has seven LEDs, including one power indicator, four FPGA control indicators, and two panel indicators. When the development board is powered on, the power indicator lights up. The four user LEDs and two panel indicators are connected to the I/O pins of the FPGA BANK65 and BANK66 . Users can control their on/off states via a program. When the I/O voltage connected to a user LED is low, the user LED is off; when the I/O voltage is high, the user LED is on. A schematic diagram of the user LED hardware connection is shown in Figure 2.10.1 .

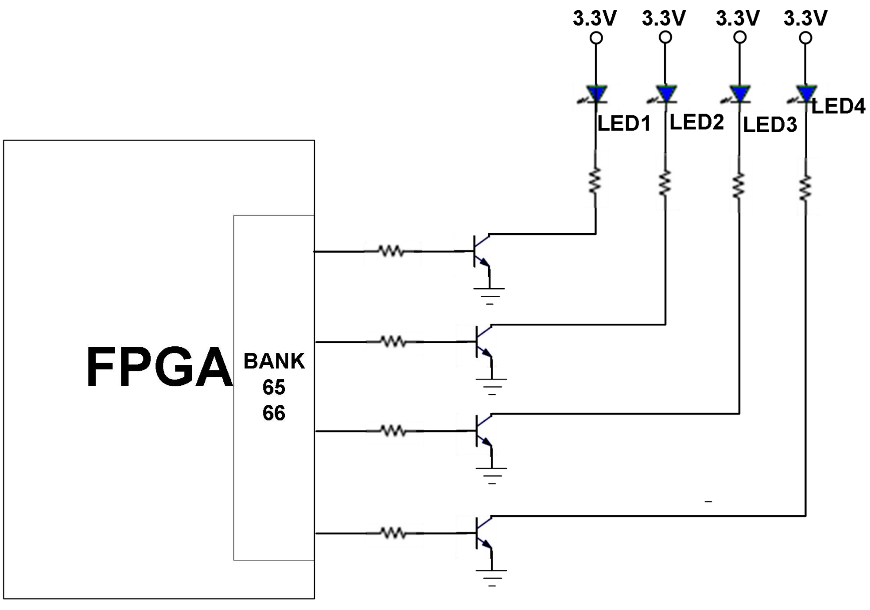

Figure 2.10.1 - Schematic diagram of user LED hardware connection

LED lamp pin assignment

Signal name	FPGA pin names	FPGA pin numbers	Remark
LED1	B66_T3U	E12	User-defined indicator lights
LED2	B66_T2U	F12	User-defined indicator lights
LED3	B66_T1U	L9	User-defined indicator lights
LED4	B65_T0U	H23	User-defined indicator lights
TEST_LED1	B66_L22_N	E13	Panel indicator lights
TEST_LED2	B66_L22_P	F13	Panel indicator lights

Table 2.10.1 - LED Pin Assignment

2.11 Buttons

The AXKU062 development board has two buttons: a reset button and a user button. Both the user button and the reset button are connected to the FPGA BANK65's I/O pins. The user button is active low and enables certain board functions for the user; the reset button is used for system reset. A connection diagram of the buttons is shown in Figure 2.11.1 .

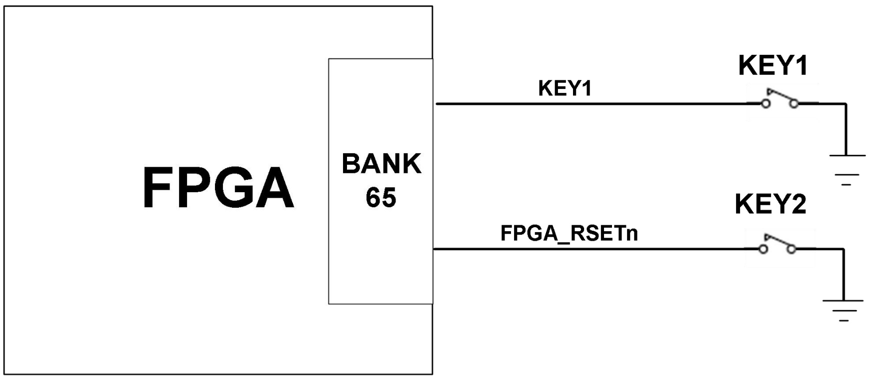

Figure 2.11.1 - Button Connection Diagram

FPGA pin assignment for the buttons:

Signal name	FPGA pin names	FPGA pin numbers	Remark
KEY1	B65_T1U	N23	User buttons
FPGA_RSETN	B65_T2U	N27	System Reset

Table 2.11.1 - Key Pin Assignment

2.12 JTAG Debug Port

The AXKU062 development board has a reserved JTAG interface for downloading FPGA programs or burning programs to FLASH. To prevent damage to the FPGA chip from hot-plugging, we added a protection diode to the JTAG signal to ensure that the signal voltage is within the acceptable range for the FPGA, thus avoiding damage to the FPGA.

2.13 Fan

Because the FPGA generates a lot of heat during normal operation, we added a heatsink and fan to the board to prevent the chip from overheating. The fan is controlled by the FPGA chip, with the control pin connected to the BANK48's I/O pin. If the I/O output is low , the MOSFET is turned on, and the fan operates.

Fan control pin assignment:

Signal name	FPGA pin names	FPGA pin numbers	Remark
FAN_PWM	B64_T0U	AK11	Fan control pin

Table 2.13.1 - Fan Pin Assignment

2.14 Power Supply

The development board's power input voltage is DC 12V. It can be powered by an external +12V power supply or via PCIe. When using an external power supply, please use the power supply that came with the development board; do not use other types of power supplies to avoid damaging the board. The power supply design diagram on the board is shown in Figure 2.13.1 below :

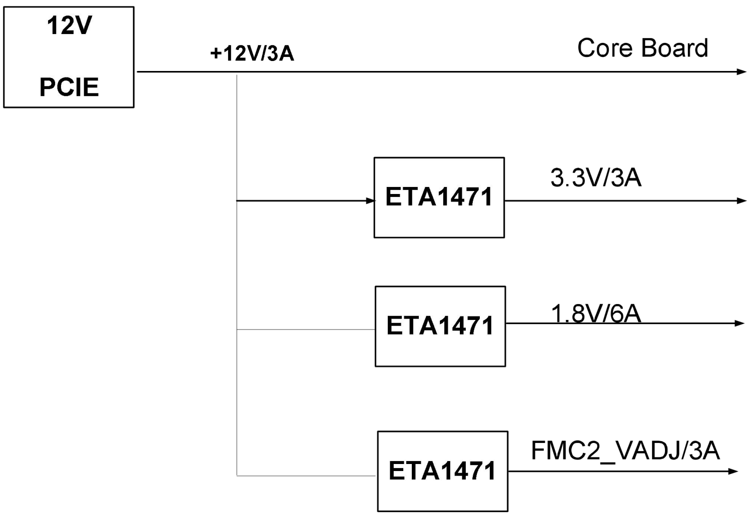

Figure 2.13.1 - Power interface section in the schematic diagram

2.15 Structural Dimensions Drawing

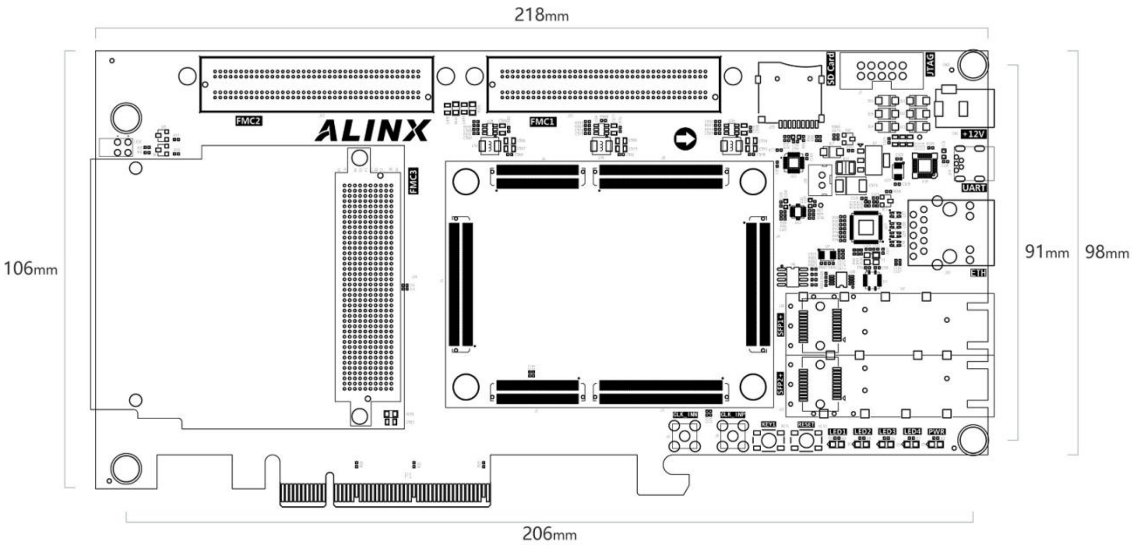

Table 2.15.1 - Top View

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